

Spatial Protection Spillovers and the Equilibrium of Crime and Policing*

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Abstract

Spatial heterogeneity in crime suggests that location is a key choice variable in criminal behavior and police allocation. To capture this, we develop a topological model in which criminals and the police interact strategically across space. The key departure from the existing literature is that distance matters: the coverage radius of a police unit generates spatial spillovers, so a location can receive protection from nearby patrols. This is in sharp contrast with discrete spatial models, where each discrete partition is isolated from its surroundings, biasing the equilibrium police allocation. We analytically characterize this bias and, more generally, demonstrate how spillovers affect the equilibrium police allocation. In particular, as spillovers increase: (i) more (fewer) units are allocated at local maxima (minima) of crime; (ii) the location where units are more concentrated shifts toward the steepest side of the crime distribution; and (iii) the entropy of the police spatial density decreases. Jointly, our results point to the essential role of spatial spillovers in the equilibrium determination of crime and police allocation.

JEL Codes: C72; K42; R12; C65

Keywords: spatial crime pattern; spatial policing strategy; spillover effects; apprehension probability

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1 Introduction

The expected profit from crime depends on several factors—such as the probability of finding a victim, the expected bounty, luminosity conditions, and local social dynamics. These factors are highly heterogeneous across space, even within the same city. As a result, crime incidence shows well-documented spatial variation (see, e.g., Weisburd 2015, Lee et al. 2017, Gill et al. 2017). At the same time, the spatial pattern of crime cannot be studied in isolation from police strategy, as crime and law enforcement interact in intricate ways. For example, empirical evidence suggests that criminals respond to an increase in police presence in a region by committing less crime there and more elsewhere—a phenomenon known as the displacement effect, documented by Di Tella and Schargrodsky (2004) and more recently by Maheshri and Mastrobuoni (2021) and Blattman et al. (2021). Such responses may, in turn, prompt the police to reallocate their units, illustrating the mutually adaptive behavior of criminals and law enforcement.

While the empirical literature has recently made substantial progress in understanding the spatial relationship between police and crime (as illustrated in Adda et al. 2014, Collazos et al. 2021, Khanna et al. 2025), theoretical work has lagged behind. Frameworks that consider the spatial nature of criminal behavior (such as Freeman et al. 1996, Lazzati and Menichini 2016, Galiani et al. 2018) typically divide the city into finite geographical cells—which may, for example, represent neighborhoods—and model criminals as choosing the cell in which to commit a crime within this discrete-choice space. By treating cells as isolated choice alternatives, this modeling approach imposes no topological structure on the city—proximity between locations plays no role—and, therefore, it abstracts from genuine spatial interaction.

The contribution of this paper is twofold. First, we demonstrate that this abstraction is not innocuous. Applying discrete-choice models to inherently continuous spaces leads to two important theoretical pitfalls: (a) equilibrium outcomes depend on how the city is partitioned, and (b) the equilibrium police allocation cannot be determined without bias. Second, to overcome these issues, we develop a tractable, empirically grounded model with an explicit topological structure.

To identify the problems of discrete models, we start by considering the type of model used in the literature. In these models, police protection in a cell depends on a public safety production function (PSPF) that links police expenditures to the probability of clearing crimes in that cell. Given the PSPF, the police decide how to distribute

their units across cells to maximize the number of crimes cleared subject to a budget constraint: increasing protection in one cell means reducing it elsewhere. Criminals, in turn, choose a geographical cell to maximize expected profits based on local gains from crime—which vary exogenously across cells—and the local probability of being caught—which depends endogenously on the spatial allocation of the police.

We show that this type of framework fails to satisfy an internal consistency property, which we term *consistent aggregation*. When equilibrium outcomes—such as the total crime rate and clearance rate—do not depend on the way the city is arbitrarily partitioned, we say that the model satisfies consistent aggregation. We find that PSPFs used in previous studies (such as Ehrlich 1973, Merlo 2003, Lazzati and Menichini 2016, Fu and Wolpin 2018) do not satisfy this property. This is problematic because the partition of the city is neither a primitive of the model nor a feature of agents’ beliefs, yet it determines equilibrium outcomes.

Beyond this internal consistency issue, we also argue that all discrete-choice models share a common limitation: they ignore spatial spillover effects arising from police patrols that provide protection beyond a single cell. In a discrete framework, a cell without assigned patrols receives no protection, even if neighboring cells are heavily patrolled, which is particularly unrealistic when cells are small. Therefore, discrete models face an inherent trade-off between spatial resolution and bias. When cells are large, spillovers are less significant, but police deployment and crime rates can only be characterized in aggregated terms over broad areas. Achieving a finer spatial description requires smaller cells, which amplifies the very bias it introduces: as cells shrink, ignored spillovers become more significant and the equilibrium police allocation predicted by the model becomes increasingly distorted.

To address these limitations, we develop a model of crime and surveillance that departs from the discrete literature in two key respects. First, the model is continuous, so criminals decide at which point they commit crimes and the police command chooses the spatial density of police units at each point of the city. To model the locations of crimes and police units, we treat them as point processes; to gain tractability, we assume the number of police units is Poisson-distributed. Second, the model is topological, so the probability of clearing a crime increases with the proximity of police units to the crime. Consequently, a location can receive police protection from patrols in its vicinity, not only from units deployed at that exact point, and the protection offered by a police unit may become less effective when it overlaps with that of other units. We obtain a closed-form equilibrium characterization for the spatial distribution of crimes and police

units, the crime rate, and the clearance probability, as a function of the primitives.

Our second main contribution is to show that, because the interaction between police and criminals is microfounded, our topological model solves these two problems simultaneously. Consider first the lack of consistent aggregation. The characterization of our model provides a relation between police expenditures and the probability of clearing crimes in the city—in practice, a PSPF. We prove that, if a city is partitioned into cells and this PSPF is used within each cell, aggregate equilibrium outcomes—such as the crime rate and the proportion of cleared crimes—no longer depend on how the city is partitioned. In this case, we say that the discrete model is induced by the PSPF of our model. The fact that our model can supply discrete models with a PSPF that ensures internal consistency is not a coincidence: since the continuous model defines police allocation pointwise and aggregates via integration, consistent aggregation holds automatically—it is not an additional requirement but a structural property of the framework.

We then turn to the second issue and study the consequences of introducing police protection spillovers. More specifically, we compare the police allocation generated by our continuous model to that of a discrete model induced by the PSPF of our model (or, equivalently, to a topological model when the coverage radius of the police goes to zero), to quantify the bias introduced by ignoring spillovers. The bias exists as long as the coverage radius is positive, and there are important qualitative differences between the two models. To illustrate, consider Figure 1. Panel (a) shows the expected profit from a successful crime as a function of location in a one-dimensional city, where criminal opportunities are more attractive at the center and decline toward the borders. Panel (b) depicts the police allocation with (blue line) and without (purple line) spillover effects. The two distributions differ significantly: a model without spillovers predicts a unimodal police allocation that mirrors the crime rate, while incorporating spillovers yields a bimodal one.

We obtain four main analytical results on how spillover effects affect the police allocation. First, we assume that the police coverage radius is small enough for a second-order approximation. In this case, discrete models underestimate the police spatial distribution wherever this density is concave: at such points, the local unit spills more protection toward less-patrolled neighbors than it receives from them, so ignoring these net outward spillovers makes the point appear more protected than it is. The analogous result holds for convex points, where discrete models overestimate the spatial density of the police. The second result is that, as spillovers increase, the location

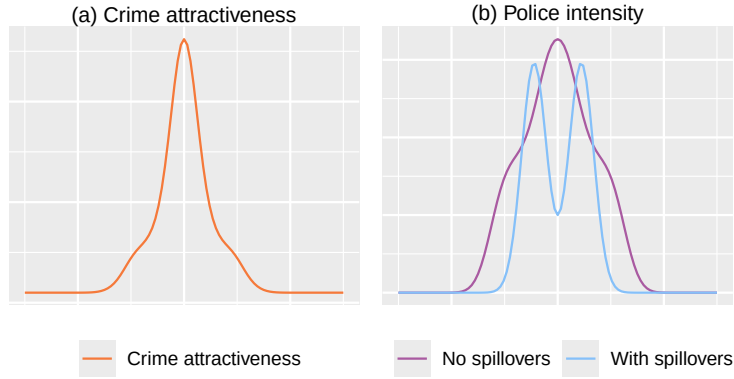


Figure 1: Panel (a) shows the crime attractiveness (i.e., expected profit from a successful crime) as a function of location $a(x) = \exp\left(\frac{\sqrt{\pi}}{12} e^{16/9} \sum_{k \in \{16, 56\}} [\operatorname{erf}(3x + \frac{k}{15}) - \operatorname{erf}(3x - \frac{k}{15})]\right)$, in a one-dimensional city $W = [-3, 3]$. Panel (b) shows the police allocation with and without spillover protection generated by our model.

where units are more concentrated shifts toward the side where the crime rate has the strongest curvature (and, thus, decreases most steeply). This occurs because on that side the local unit spills more protection outward than it receives back, so more units are needed there to compensate for the net outward loss. Third, since spillovers increase the peaks and deepen the valleys of the police allocation, its entropy—a measure of how evenly police units are spread across the city—decreases: the distribution becomes more concentrated as spillovers grow. Then, we consider a higher-order approximation to derive, in a parametrized example, the conditions under which a local maximum of the crime rate becomes a local minimum of the police spatial distribution, as in Figure 1.

Needless to say, our results rest on the assumptions of the model. The police allocation is modeled as a Poisson point process, so the number of units in the target city follows a Poisson distribution and units are independently deployed. This specification offers considerable tractability and allows the police command to set arbitrarily different amounts of police protection at different points of the city, as long as the budget constraint is satisfied. We also assume that crime and police locations are independent. This assumption captures a realistic feature that police do not know where criminals will strike, and criminals do not know where police will be deployed. However, by assuming independence, we abstract from visual deterrence effects, whereby a criminal refrains from committing a crime upon observing a police patrol. As a consequence of these assumptions, our closed-form equilibrium results are not general properties of

spatial policing models, but rather reflect a tractable decomposition obtained under a specific stochastic-geometric structure.

The rest of the paper is organized as follows. The next section lays out the general problem, notation, and properties used in both discrete and continuous models. In Section 3, we present the generic discrete framework, show how typical models used in the literature fail to satisfy consistent aggregation, and argue that discrete models cannot determine the police spatial distribution. Section 4 presents the continuous model and its equilibrium. Then, Section 5 analytically characterizes the consequences of introducing spillover effects on the equilibrium police allocation, and Section 6 concludes.

2 The General Problem

We aim to develop a spatial model that captures the interaction between police and criminals. This section describes the components common to both discrete- and continuous-choice crime models.¹ The police command operates under a budget constraint and must decide how to allocate patrols across different regions. We assume that their objective is to maximize apprehensions and minimize crime, which, in our framework, are equivalent.² The level of police enforcement in a region depends on the resources allocated by the police command. Increasing police expenditures raises the probability of clearing crimes in that region, as supported by, for example, Levitt (2002) and Evans and Owens (2007). Criminals choose where to commit crimes to maximize their payoffs, taking into account both the expected gains from criminal activity and the probability of apprehension.

All proofs of the paper are in Appendix B. Also, at the end of the paper, we provide a summary of the notation used throughout for ease of reference.

2.1 Notation, Fundamentals, and Equilibrium Concept

To fix notation, let $W \subset \mathbb{R}^2$ be a compact set representing a police force’s jurisdiction. W has C^1 boundary and we denote its boundary by $\text{Bd}(W)$. We refer to an arbitrary two-dimensional subset $W' \subset W$ as a *region*, and its area is denoted by $|W'|$. The set of

¹Additional assumptions used exclusively in the discrete- and continuous-choice models are introduced in Section 3 and Subsection 4.1 respectively.

²Corollary 1 in Subsection 4.2 provides a formal proof of this equivalence for the continuous model we propose. We adopt the maximization of apprehensions as the police objective to streamline the exposition.

positive continuous functions on W is denoted by $\mathcal{C}_{++}$. For any non-negative integrable function v on W such that $\int_W v(x)dx > 0$, we define f_v as the normalization of v , that is, $f_v := v / \int_W v(x)dx$. We define δ as the Dirac delta function, so that $\int_{W'} \delta(x - x_0)dx = 1$ if $x_0 \in W'$ and 0 otherwise.

We next describe the fundamentals of this game. Law enforcement determines that the police command operates under a budget Λ and applies a punishment b to a criminal when a crime is cleared. We consider a pool of N risk-neutral potential criminals who contemplate the possibility of committing crimes. Each criminal has an opportunity cost drawn from a cumulative distribution function (CDF) F_κ , strictly increasing, to engage in criminal activity. By incurring this cost, an agent becomes a criminal and selects a location $x \in W$ in which to commit the crime.

We also define the criminals' attractiveness function $a \in \mathcal{C}_{++}$. This function is interpreted as follows: if a criminal can choose a point $x \in W$ to commit a crime (as in a continuous-choice model), $a(x)$ represents the expected monetary gain if the crime is not cleared; if the criminal can choose a region $W' \subset W$ to commit a crime (as in a discrete-choice model), the expected monetary gain if the crime is not cleared is the average of a over W' .³ We assume that a is infinitely differentiable. Figure 2 illustrates an example of the attractiveness function in a city $W = [0, 1] \times [0, 1]$. The colors in the figure represent the attractiveness at each point within the city. We can see two points where the attractiveness is maximal.⁴

Motivated by empirical evidence suggesting that criminals and police learn about each other's preferences through repeated interactions (see, e.g., Knowles et al. 2001), we adopt the Nash equilibrium with rational expectations as our solution concept. Thus, although the game is formally static, the Nash equilibrium captures an implicit dynamic interaction between criminals and the police. Variables with a superscript $\star$ denote

³Despite studies on the statistical relation between neighborhood characteristics and crime patterns dating back to the 19th century (see Guerry 1833 and Quetelet 1842), the determinants of crime distribution are disputed in the criminology literature. Therefore, we deliberately refrain from providing a microfoundation for it—we only require that the criminals' preferences exist and are well-defined. For example, while some empirical papers find that the density of potential victims attracts crime (Brantingham and Brantingham 1995, Harries 2006, Cabrera-Barona et al. 2019), others find that crime is attracted to a weaker social cohesion (Shaw and McKay 1942, Sampson and Groves 1989, Bernasco and Block 2009), and yet others find that criminals prefer to commit crimes in regions similar to their neighborhood (Chamberlain and Boggess 2016).

⁴The attractiveness level at $(x, y) \in W$ is given by

$$a(x, y) = \exp(-\alpha_0((x - x_0)^2 + \alpha_1(y - y_0)^2)) + \exp(-\alpha_2((x - x_1)^2 + \alpha_3(y - y_0)^2)).$$

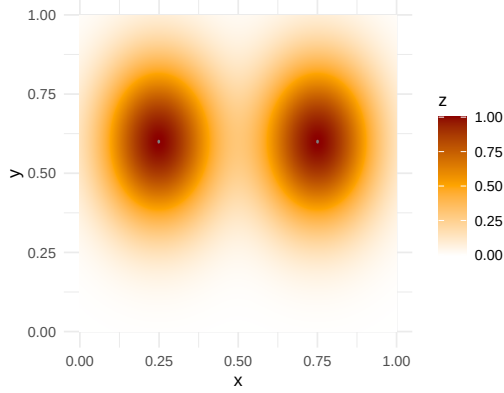


Figure 2: Example of attractiveness level within $W = [0, 1] \times [0, 1]$

equilibrium values.

3 Drawbacks of Discrete Frameworks

In this section, we show that discrete-choice models suffer from a problem of internal consistency in their most general formulation and that they introduce a bias when used to determine the equilibrium police location.

Consider a partition of W , given by $\{W_i\}_{i \in I}$, where $I \subseteq \mathbb{N}$ and $|W_i| > 0$. The police can allocate a portion of the budget to each region, and each criminal decides whether and where to commit a crime choosing an element of I . If a crime in W_i is not cleared, the criminal earns a profit equal to the average attractiveness of W_i , given by $a(W_i) := \frac{1}{|W_i|} \int_{W_i} a(x) dx$ for each $W_i \subseteq W$. If we set $a(\{x\})$ as the limit of sets with positive areas converging to x , we have $a(\{x\}) = a(x)$.⁵

The critical modeling decision is how to set the public safety production function (PSPF): the probability of clearing a crime in each region W_i as a function of the expenditures on police in W_i , denoted by Λ_i . We impose that this function depends only on the fundamentals of W_i (the shape and the function a inside it), that it is bounded in $[0, 1]$, strictly increasing, and strictly concave in Λ_i . We denote the clearance probability of region W_i as $\pi_{W_i}(\Lambda_i)$ for each $\Lambda_i \in \mathbb{R}_+$.

The equilibrium arising from this model is standard in the literature and is presented in Appendix A. If W' is union of a non-empty subset of $\{W_i\}_{i \in I}$, we denote by $M_{W'}^*$ the

⁵Formally, if $(W'_n)_n$ is a sequence of measurable sets with $|W'_n| > 0$ for each n converging to a point $x \in W$ with respect to the Hausdorff distance, and we define $a(\{x\}) := \lim_{n \rightarrow \infty} a(W'_n)$, then $a(\{x\}) = \int_W a(x') \delta(x' - x) dx' = a(x)$.

equilibrium expected number of crimes, by $p_{W'}^*$ the equilibrium expected proportion of crimes that are cleared, and by $\Lambda_{W'}^*$ the equilibrium police expenditures in W' .

3.1 Inconsistency in Discrete Models

The consistency problem of discrete-choice models used in the literature is that they cannot allow for what we term consistent aggregation. To define it formally, let $I' \subset I$ be non-empty and define the aggregate clearance rate (ACR) of $\bigcup_{i \in I'} W_i$ under the partition $\{W_i\}_{i \in I'}$ as

$$p_{\{W_i\}_{i \in I'}}^* := \frac{\sum_{i \in I'} M_{W_i}^* p_{W_i}^*}{\sum_{i \in I'} M_{W_i}^*}.$$

Note that when the partition consists of a single set, $\{W_i\}_{i \in I'} = \{W_i\}$ and $p_{\{W_i\}}^* = p_{W_i}^* = \pi_{W_i}(\Lambda_{W_i}^*)$. The consistent aggregation principle states that equilibrium outcomes cannot depend arbitrarily on the partition of the police jurisdiction W .

Consistent Aggregation Principle. *We say that a model satisfies consistent aggregation when the following three conditions hold:*

- (i) *The equilibrium crime rate in W does not depend on the way the set is partitioned;*
- (ii) *The ACR of a set $\bigcup_{i \in I} W_i$ does not depend on how the regions are aggregated, so*

$$p_{\{W_i\}_{i \in I}}^* = p_{\{\bigcup_{i \in I} W_i\}}^*;$$

- (iii) *Police enforcement expenditures are additive across regions, hence*

$$\sum_{i \in I} \Lambda_{W_i}^* = \Lambda_{\bigcup_{i \in I} W_i}^*.$$

Numerical illustration of PSPF inconsistency. We consider two functional forms for π_{W_i} used in the literature to show the lack of consistent aggregation of PSPFs. Merlo (2003) considers the following specification:

$$\pi_{W_i}(\Lambda_i) = \begin{cases} 0 & \text{if } \Lambda_i \leq 1 \\ 1 - \xi \Lambda_i^{-\alpha} & \text{if } \Lambda_i > 1, \end{cases}$$

with $\xi > 0$ and $\alpha \in (0, 1)$, while Fu and Wolpin (2018) specify the function

$$\pi_{W_i}(\Lambda_i) = \exp\left(-\frac{\gamma}{\Lambda_i}\right),$$

with $\gamma > 0$, when the crime rate has no impact on the clearance probability.

We use these two specific functions and the attractiveness pattern from Figure 2 to illustrate how this approach fails to satisfy the consistent aggregation principle, as both the crime rate and the ACR implied by these models depend on how the city is subdivided. To demonstrate this, we consider four different ways of partitioning our unitary city into regions, depicted in Figure 3. Splits A and B divide the city into two areas of equal size, each being a rectangle measuring 1×0.5 , yielding an area of 0.5 units. Splits C and D subdivide the city into four units of equal area. In Split C, we obtain four squares with side length 0.5, while in Split D, the regions are isosceles triangles of area 0.25.

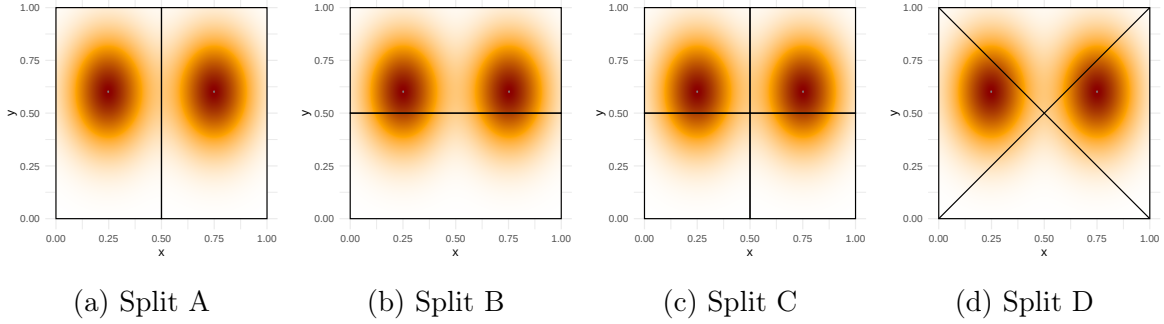


Figure 3: Example of partitions of a city

We solve numerically for the equilibrium of this economy under different splits and different functional forms for the technology of clearing crimes used by Merlo and by Fu and Wolpin. We set all parameters for the two clearance functions to yield an ACR of 0.7 in the Split A case. Table 1 shows the results. We can see that with Split A and the Merlo (2003) and Fu and Wolpin (2018) functions, we have an ACR of 0.7 (by construction) and a crime rate of 0.0152. If we solve for optimality under Split B, which also has two subregions of the same size as Split A, we would have a different result with an ACR of 0.7336 and a crime rate of 0.0282 in Merlo's model and an ACR of 0.7249 and a crime rate of 0.0151 in Fu and Wolpin's model. The table shows that these results depend on the number of splits as well as on the specific geometric format of these splits. In other words, the same discrete model of crime can yield different results

depending on the clearance technology and how the city is partitioned. Interestingly, even if the number of partitions and their areas are the same, as when comparing Split A with Split B, and Split C with Split D, the results not only differ, but they move in different directions depending on the specific PSPF.

Table 1: Aggregate Clearance Rate and Crime Rate by technology and split

Clearance Technology	Split	# Units	ACR	Total crime
Merlo (2003)	Split A	2	0.7000	0.0152
	Split B	2	0.7336	0.0282
	Split C	4	0.7336	0.0282
	Split D	4	0.7281	0.0301
Fu and Wolpin (2018)	Split A	2	0.7000	0.0152
	Split B	2	0.7249	0.0151
	Split C	4	0.7249	0.0151
	Split D	4	0.7328	0.0148

Notes: We set $\Lambda = 100$ (which induces interior solutions in all cases), $b = 0.1$, $N = 1$, F_κ as the uniform distribution on $[0, 1]$. For Merlo (2003) we have $\pi_M = 1 - \xi(\Lambda + 1)^{-\alpha}$, with $\xi = 1$ and $\alpha = 0.26$. For Fu and Wolpin (2018) we have $\pi_{FW}(\Lambda) = \exp\left(\frac{-\gamma}{\Lambda}\right)$, with $\gamma = 35.67$. As the areas in Splits A and B differ from those in Splits C and D, we compute Λ_i as the expenditure per unit of area.

Although the institutional context (e.g., the political division of the city into administrative regions or beats) can lead to equilibria that are partition-dependent, this sensitivity is generally problematic for three reasons. First, even in models that abstract from any institutional context (i.e., the division of the city into regions is not a primitive), a partition must be chosen, making the sensitivity an unavoidable feature of the approach, not a specific institutional artifact. Second, even if the granularity of the partition reflects a technology or behavioral constraint, there are infinitely many ways of drawing partitions with the same granularity. Third, the way the equilibrium depends on the partitions cannot be anticipated since it depends on the way the PSPF interacts with the partition and the PSPF itself is an arbitrary modeling choice.

3.2 Indeterminacy in Police Allocation in Discrete Models

In addition to the problem of consistency, discrete-choice frameworks cannot determine the equilibrium allocation of patrols within a cell. A natural approach to deal with this problem is to refine the spatial partition by making cells arbitrarily small, but

this creates an additional issue. When cells are small, patrols assigned to a given cell are likely to provide protection to neighboring cells. By treating each cell in isolation from the others, discrete models thus fail to capture spillover effects across regions.

Therefore, discrete models face an inherent trade-off. If the jurisdiction is partitioned into a small number of large cells, the resulting outcomes are silent on what happens inside each cell, providing an incomplete characterization. If the partition generates a large number of small cells, it improves the characterization of actual police deployment and crime rates, but it abstracts away from spillovers and how the small cells interact.

Analytically, in the absence of spillover effects, the equilibrium spatial distribution of police patrols depends pointwise on local crime attractiveness. That is, in discrete-choice models, if the attractiveness at $x \in W$ and $x' \in W$ satisfies $a(x) = a(x')$, then expenditures on police at x and x' (and, consequently, the number of patrols around these points) are necessarily identical regardless of the attractiveness in the surrounding areas of x and x' . The next section shows how a continuous model departs from this pointwise dependence and allows the spatial distribution of patrols to depend on the entire attractiveness function.

4 A Continuous Framework

In this section, we develop a continuous-choice model in which the level of enforcement depends on the actual distance between the location of a crime and police patrols. This approach is appealing for several reasons.

First, our model relies on empirically grounded assumptions. Evidence suggests that reducing the distance between patrol units and the crime location increases the likelihood of clearing the crime (see Weisburd 2021, Mastrobuoni 2019, and Blanes i Vidal and Kirchmaier 2018). As Blanes i Vidal and Kirchmaier emphasize, while patrol officers rarely apprehend suspects on site, closer proximity improves the preservation of material evidence and enhances the probability of identifying witnesses with relevant information.

Second, the model delivers the equilibrium distribution of crime and patrols at each point, not merely their aggregate levels, as discussed in Subsection 3.2. Also, in our model, depending on the coverage radius of each patrol, the density of units may not align with areas of higher criminal activity—a feature discrete-choice models cannot capture.

Third, the model provides a microfoundation for the clearance mechanism, rather than modeling clearance probabilities in reduced form, as is common in the literature. By considering the maximal level of spatial disaggregation of W —with all points admissible as both crime locations and sites of police protection—our model generates well-behaved equilibrium objects.

Finally, the model is both analytical and flexible, enabling a clearer understanding of the strategic interaction between criminals and the police, as well as the policy implications.

Before presenting the formal model, we highlight key differences relative to discrete-choice frameworks. In our continuous-choice model, the police command determines the probability of patrol presence at each point of W as perceived by criminals. The probability of clearing a crime at location x depends on the proximity of patrols, so each point in W is associated with a distinct clearance probability. Given that each point in W is characterized by a probability of crime occurrence and a probability of clearance, the police’s total payoff—the utility of the police command—is obtained by integrating the product of these two quantities over W .

As before, each criminal decides whether to enter, but now chooses a point $x \in W$ at which to commit a crime. If the crime occurs at x , the criminal receives a payoff $a(x)$ if it is not cleared and incurs a penalty b otherwise. The probability of clearance at x depends on the spatial distribution of police patrols, and each potential criminal maximizes their expected payoff.

4.1 Model Set-up

In what follows, we formally describe the main components of the model.

The Clearance Mechanism

To capture the dependence of the clearance probability on the proximity of patrols, we introduce a *crime-detecting kernel* K . If x denotes the location of a crime and y the location of a police patrol, the probability that the patrol detects the crime is given by $K(\|x - y\|)$. The function K is non-increasing and satisfies $K(\rho) = 0$ for $\rho > r$ and we call r the coverage radius. We assume that r is small relative to the overall dimensions of the city or region where crimes may occur, so that border effects can be neglected.

The simplest kernel is the uniform one, where $K = \beta \mathbb{1}_{\{(\cdot) \leq r\}}$ for $\beta \in [0, 1]$. In this case, a patrol detects a crime with probability β if the distance between the crime and

the patrol is less than or equal to r , and with probability 0 if the distance exceeds r . Figure 4 illustrates three examples of different crime-detecting kernels.

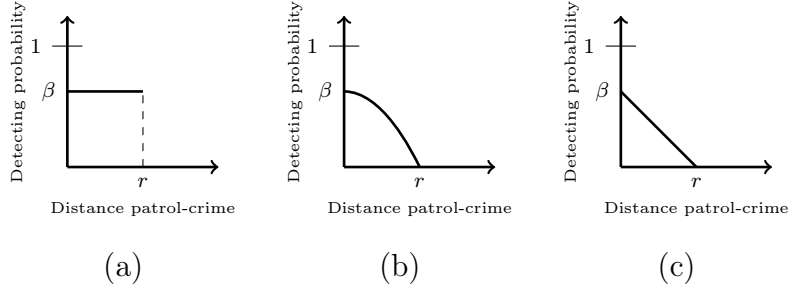


Figure 4: Different crime-detecting kernels. Panel (a) presents a uniform kernel, Panel (b) a parabolic one, and Panel (c) a linear one. In all cases, the maximum detection probability is β (potentially equal to one) when the crime occurs at the location of the police patrol, and this probability decreases to 0 as the distance between the patrol and the crime exceeds r .

Police patrols detect crimes independently, and a crime occurring at location x is cleared if at least one police patrol detects it.⁶ Consequently, if, for example, four police patrols are near x , the probability of clearing the crime is typically higher than if only a single patrol were present. This reflects the idea that areas with greater police presence have a higher likelihood of crime clearance.

Therefore, if there are N_p patrols and the set of points $\{y_1, y_2, \dots, y_{N_p}\}$ represents the locations of police patrols, the probability that a crime at $x \in W$ is cleared is

$$1 - \prod_{j=1}^{N_p} (1 - K(\|x - y_j\|)).$$

Figure 5 shows the clearance probabilities as a function of the locations of crimes and police patrols with a uniform crime-detecting kernel. Darker areas correspond to higher clearance probabilities.

Informational Structure and Point Processes

Since the police command does not know the exact locations of crimes in advance, crime locations are random from the police perspective, and we model them as a point

⁶The key idea is that proximity to patrols increases the probability of clearance. One could also assume that when the distance between a patrol and a crime exceeds r , there remains a positive constant probability of clearance (we assume this constant is zero). Incorporating this assumption does not affect our main results and would align more closely with Blanes i Vidal and Kirchmaier (2018) (see Figure 4 in their paper).

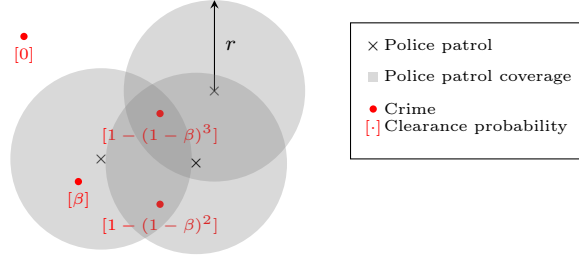


Figure 5: Clearance probability as a function of the locations of patrols and crimes within W under a uniform crime-detecting kernel. A patrol located within a radius r of a crime detects it with probability β .

process Φ . A point process can be viewed as a random element, analogous to a random variable, but taking values in the space of locally finite point configurations. While the probability law of a real-valued random variable Z determines, for example, the probability that Z falls within the interval $[0, 1]$, the probability law of a point process Φ specifies the probability that, for instance, no more than three points of Φ fall within a given region $A \subset W$. The distribution of Φ is determined endogenously by the equilibrium, and we do not, in principle, rule out the theoretical—albeit unrealistic—possibility that Φ follows a degenerate distribution, resulting in a deterministic set of crime locations.

One of the properties of a point process is its *intensity*. We denote the intensity of Φ by μ , which characterizes the surface density of crimes. That is, while Φ represents the record of exact locations of crimes, μ can be seen as the heat map of crimes, indicating the areas where crimes are more likely to occur. More precisely, the quantity $\mu(x) \Delta x$ can be interpreted both as the probability of a crime occurring in a small area Δx around x and as the expected number of crimes around x . Intuitively, the expected number of crimes occurring around the points $\{x', x''\}$ is $\sum_{x \in \{x', x''\}} \mu(x) \Delta x$. Similarly, the expected number of crimes occurring in a set A , $\mathbb{E} \left[\sum_{x \in \Phi} \mathbb{1}_{\{x \in A\}} \right]$, is given by $\sum_{x \in A} \mu(x) dx = \int_A \mu(x) dx$, and $\int_W \mu(x) dx$ is the total expected number of crimes.

This relation formally holds and, more generally, if $f : \mathbb{R}^2 \rightarrow \mathbb{R}_+$ is a measurable function, m is the number of criminals, and $\Phi = \{x_1, x_2, \dots, x_m\}$, we have that

$$\mathbb{E}[f(x_1) + f(x_2) + \dots + f(x_m)] = \mathbb{E} \left[\sum_{x \in \Phi} f(x) \right] = \int_{\mathbb{R}^2} f(x) \mu(x) dx,$$

and we refer to this result as the *intensity property* (see Baddeley 2007).

Analogously, as long as potential criminals do not know the exact locations of police patrols in equilibrium, the set of patrol locations appears random from the criminals'

viewpoint. Thus, we model patrol deployment as a point process Ψ , satisfying properties that will be defined later. We denote by λ the intensity of Ψ , or the *police intensity*, and $\lambda(x) \Delta x$ represents the probability of finding a patrol in a small region around x with area Δx . Thus, λ is to the police what μ is to crime. The randomness of Ψ does not imply that the police command is unaware of patrol locations; rather, it reflects the strategic intention of conducting preventive patrols, making the presence of police appear unpredictable to potential criminals.

Since police patrols generate spatial spillovers of protection, a location receives protection from nearby units, not only from patrols deployed exactly there. Thus, we define the *police activity* at x under police intensity λ as $z(x) := \int_W K(\|x - y\|) \lambda(y) dy$ for each $x \in W$. That is, $z(x)$ is the average number of police patrols that would detect a crime occurring at location x . Note that police protection and police intensity can, in theory, differ substantially: x might be close to several police units (thus, $z(x)$ is high) even if the likelihood of finding patrols at x is zero (thus, $\lambda(x) = 0$).⁷ For well-behaved kernels, z pins down λ uniquely.

Because the police command deploys patrols before knowing the locations of crimes, and criminals choose crime locations without knowing the locations of patrols, Ψ and Φ are independent point processes. Here and in the following, *independent* refers to probabilistic independence, which does not prohibit, for example, the police from favoring patrol deployment in locations with a high intensity of crimes.

Since the theory of point processes is not standard in Economics, we illustrate a few examples that could characterize Φ , along with their associated distribution laws and intensity measures.

Example 1. Suppose that there is exactly one crime uniformly distributed over W (this is known as a Binomial point process). Then, it is easy to see that the probability law of Φ is described as: for any $A \subset W$, $\Pr((\# \text{ crimes in } A) = 0) = 1 - |A|/|W|$. In addition, the intensity of Φ in this case is given by $1/|W| \times \mathbb{1}_{\{x \in W\}}$. That is, the probability that the crime is around $x \in W$ is $1/|W| dx$. Observe that the expected number of crimes in W is $\int_{\mathbb{R}^2} 1/|W| \times \mathbb{1}_{\{x \in W\}} dx = 1$. $\square$

Example 2. Suppose that the number of crimes in W follows a Poisson distribution with mean 1, and the location of each crime event is independently and uniformly distributed over W (this is known as a Poisson point process). Then it is possible to show

⁷This is precisely the case of Figure 1.

that the probability law of Φ is fully described as: for any $A \subset W$, $\Pr((\# \text{ crimes in } A) = 0) = e^{-|A|/|W|}$. As in the previous example, the intensity of Φ is $1/|W| \times \mathbb{1}_{\{x \in W\}}$ (i.e., the intensity alone does not completely characterize a point process).⁸ The expected number of crimes in W is $\int_{\mathbb{R}^2} 1/|W| \times \mathbb{1}_{\{x \in W\}} dx = 1$. $\square$

Example 3. Suppose that Φ is deterministic and there are exactly three crimes in W located at x_1 , x_2 , and x_3 , respectively. In this case, $\Pr(\Phi = \{x_1, x_2, x_3\}) = 1$. The intensity is given by $\delta(x - x_1) + \delta(x - x_2) + \delta(x - x_3)$. The expected number of crimes in W is $\int_{\mathbb{R}^2} (\delta(x - x_1) + \delta(x - x_2) + \delta(x - x_3)) \mathbb{1}_{\{x \in W\}} dx = 3$ since $x_1, x_2, x_3 \in W$. $\square$

The Police Command

We assume the following two properties for the locations of police patrols, represented by Ψ . First, the number of patrols available to the police command is modeled as a Poisson random variable with exogenous mean Λ , which can be interpreted as a function of police expenditures determined by political agents. The uncertainty in this number reflects the bureaucratic, legal, and logistical constraints that affect the actual deployment of police units. In addition, since the standard deviation of the patrol count, $\sqrt{\Lambda}$, is small relative to its mean when Λ is large, the number of patrols can be approximated by Λ with a small relative deviation. Therefore, the results are statistically similar to those obtained under a fixed (deterministic) number of patrols.

Second, the police command deploys each unit so that, from the criminals' viewpoint, each unit's location is drawn from a common distribution over W —given by $\lambda(x)/\int_W \lambda(x') dx'$ if the intensity of Ψ is λ . Although this places a restriction on the strategy set available to the police command, this environment still supports a wide range of policing strategies. Indeed, provided the budget constraint is satisfied, it allows the police command to achieve any desired level of local enforcement *ex ante* (i.e., any arbitrary value of intensity at some point $x \in W$). In addition, from the police command's perspective, the precise *ex post* locations of police patrols are irrelevant, as it aims to maximize its *expected* payoff.

The two properties of Ψ , taken together, imply that the set of patrol locations follows a Poisson point process (PPP). Moreover, because the distribution of each patrol is

⁸To grasp the intuition, observe that when $|A|$ is small we have that $\Pr((\# \text{ crimes in } A) > 1) \ll \Pr((\# \text{ crimes in } A) = 1)$ because it is unlikely that two points fall in A . This means that $\Pr((\# \text{ crimes in } A) = 1) \approx \Pr((\# \text{ crimes in } A) > 0) = 1 - e^{-|A|/|W|} \approx |A|/|W|$. When $A \subset W$ tends to a point with area dx , we have that $\Pr((\# \text{ crimes in } A) = 1) = 1/|W| dx$.

continuous, the intensity measure of the police is in $\mathcal{C}_{++}$. A notable characterizing property of the PPP is that the numbers of points falling in disjoint regions of W are independent and Poisson-distributed. Loosely speaking, what happens at a location in W (for example, whether or not there is a patrol) does not affect what happens in other regions of W . This confers analytical tractability, ease of implementation, and good statistical behavior on PPPs.⁹ The properties of PPPs we use in this paper are detailed in Appendix C.

Now we characterize the objective function of the police command. Each cleared crime increases the police command's payoff by one unit. Therefore, the expected utility of the police command is given by

$$U^{\text{pol}}(\lambda, \mu) := \mathbb{E} \left[\sum_{x \in \Phi} \left(1 - \prod_{y \in \Psi} (1 - K(\|x - y\|)) \right) \right],$$

where the intensities of Φ (crime locations) and Ψ (patrol deployment) are given, respectively by μ (crime intensity) and λ (police intensity). By using the properties of point processes, we can show that

$$U^{\text{pol}}(\lambda, \mu) = \int_W \mu(x) \left(1 - e^{-\int_W K(\|x-y\|) \lambda(y) dy} \right) dx. \quad (1)$$

Appendix B details the steps to derive (1). The term $\mu(x)dx$ is the maximum expected payoff the police command can receive for protecting the point x , while the term in parentheses represents the expected proportion of crimes around x that are cleared for a given patrolling intensity λ . Therefore, the police command seeks to maximize the right-hand side of (1) over $\lambda \in \mathcal{C}_{++}$, subject to the constraint $\int_W \lambda(x) dx \leq \Lambda$. Note that no assumptions about Φ or its intensity μ are required to obtain (1).

⁹PPP's also have nice dynamic properties. A PPP can be seen as the equilibrium state of a Brownian-based diffusion. Consider an initial configuration of patrols distributed as a PPP with intensity λ . Assume that each point is displaced according to a Brownian motion with a drift $\nabla \ln(\lambda)$, where ∇ is the gradient operator. Ignoring border effects, as the patrols move, their distribution remains that of a PPP, preserving the initial intensity λ . This dynamic process could mirror some randomness in the movement of patrols, with the drift ensuring that the patrols provide the protection intensity chosen by the police command. The observed pattern of patrols is interpreted as a static snapshot of the dynamic process described above. See Stoyan et al. (1987) for more details on PPPs.

Criminals

As in the discrete-choice model, the number of potential criminals is N , each agent faces an idiosyncratic opportunity cost of committing a crime, drawn from a distribution with CDF F_κ , and the expected bounty from a successful crime at x is $a(x)$.¹⁰ In addition to the entry decision, each potential criminal chooses a pure or mixed strategy over locations in W at which to commit a crime. We denote by $\mathcal{S}$ the space of distributions over W .

Now we write the utility of criminals as a function of the police intensity λ and an arbitrary distribution s over W , representing the criminal's strategy. Let X be a random location drawn from s . Then, the expected payoff from committing a crime under this strategy is given by

$$U^{\text{crim}}(\lambda, s) := \mathbb{E} \left[a(X) \prod_{y \in \Psi} (1 - K(\|X - y\|)) - b \left(1 - \prod_{y \in \Psi} (1 - K(\|X - y\|)) \right) \right].$$

Using the properties of point processes and denoting the *adjusted attractiveness* by $\tilde{a} := a + b$, we obtain

$$U^{\text{crim}}(\lambda, s) = \int_W \tilde{a}(x) e^{-\int_W K(\|x-y\|) \lambda(y) dy} s(x) dx - b. \quad (2)$$

We provide the derivation of this expression in Appendix B. Note that the criminal's problem is equivalent to one with no punishment, where the spatial attractiveness is replaced by the adjusted attractiveness. This happens because a criminal who successfully commits a crime earns b from avoiding punishment, in addition to the crime profits a .

Formally, a potential criminal with an opportunity cost κ does not commit a crime if $\max_{s \in \mathcal{S}} \{U^{\text{crim}}(\lambda, s)\} < \kappa$, and becomes a criminal if $\max_{s \in \mathcal{S}} \{U^{\text{crim}}(\lambda, s)\} > \kappa$. When this last condition is satisfied, the support of the mixed strategy used by the criminal lies in the set $\text{argmax}_{x \in W} \{\tilde{a}(x) e^{-\int_W K(\|x-y\|) \lambda(y) dy}\}$.

4.2 Equilibrium

We proceed to characterize the equilibrium. Formally, it is given by a police intensity λ^* and a criminal intensity μ^* , such that the three following conditions hold:

¹⁰For all results in this section, it is only required that the *average* number of potential criminals is N .

- (i) The police command maximizes its expected utility, so $\int_W \lambda^*(x) dx \leq \Lambda$ and $U^{\text{pol}}(\lambda^*, \mu^*) \geq U^{\text{pol}}(\lambda, \mu^*)$ for any function $\lambda \in \mathcal{C}_{++}$ with $\int_W \lambda(x) dx \leq \Lambda$;
- (ii) Each potential criminal with idiosyncratic cost κ commits a crime if, and only if, $\max_{s \in \mathcal{S}} \{U^{\text{crim}}(\lambda^*, s)\} > \kappa$ and each criminal has a location strategy in the set $\text{argmax}_{s \in \mathcal{S}} \{U^{\text{crim}}(\lambda^*, s)\}$;
- (iii) The individual and aggregate behavior of potential criminals is consistent—in particular, the expected number of crimes is $\int_W \mu^*(x) dx$.

Before analyzing the equilibrium behavior of each type of agent, we introduce some useful notation. First, for normalized densities f_1 and f_2 with support in W , we denote the entropy of f_1 by $\mathcal{H}(f_1) := -\int_W f_1(x) \ln f_1(x) dx$, the cross-entropy between f_1 and f_2 by $\mathcal{H}(f_1, f_2) := -\int_W f_1(x) \ln f_2(x) dx$, and the relative entropy between f_1 and f_2 (also known as the Kullback–Leibler divergence) by $D(f_1 \| f_2) := \int_W f_1(x) \ln(f_1(x)/f_2(x)) dx$. Note that $\mathcal{H}(f_1, f_2) = D(f_1 \| f_2) + \mathcal{H}(f_1)$. We also define the Fisher information of f_1 by $\mathcal{I}(f_1) = \int_W \|\nabla f_1(x)\|^2 / f_1(x) dx$. Finally, we denote by u the uniform distribution over W and by U a uniform random variable.

Second, we let $q := \int_{\mathbb{R}^2} K(\|x\|) dx$ be the effective coverage of each police patrol, a measure of the technology quality to detect crimes, and $C := \Lambda q / |W|$ be the police capacity, which depends on the total budget, technology, and area to cover.

Additionally, for any function $f \in \mathbb{R}_+^W$, define the constant

$$\underline{C}(f) := \frac{\int_W \ln(f(x)) dx}{|W|} - \ln \min\{f\}.$$

To simplify the exposition, we assume that the police capacity satisfies $C > \underline{C}(\tilde{a})$. This ensures that the police cover the entire region W in equilibrium when r is small enough.¹¹ If this condition is not met, the police cover only a subset of W and all our results hold in this subset.¹²

Lastly, we define the ACR in W as in the discrete-choice model: the expected ratio of the number of crimes cleared to the number of crimes in equilibrium. Thus, denoting

¹¹The threshold is the difference between the average and the minimum of $\ln(\tilde{a})$. When this term is large, the relative attractiveness of the region around $\text{argmin}\{\tilde{a}\}$ is small, and the police command might choose to turn a blind eye to the region. The threshold of r ensuring a well-behaved solution is $r^2 < \frac{1}{\eta} \min \left\{ \frac{z^*(x)}{\nabla^2 z^*(x)} : \nabla^2 z^*(x) > 0 \right\}$ where η depends on the shape of the crime-detecting kernel, but not on the radius. This is discussed in further detail in Section 5.

¹²This subset is $\{x \in W : \tilde{a}(x) \geq \bar{a}\}$ for a well-defined $\bar{a}$.

this average as p_W^* (the same notation used previously), we have¹³

$$p_W^* = \mathbb{E} \left[\frac{\sum_{x \in \Phi} \left(1 - \prod_{y \in \Psi} (1 - K(\|x - y\|)) \right)}{\sum_{x \in \Phi} \mathbb{1}_{\{x \in W\}}} \right].$$

This quantity can be seen as a measure of police effectiveness.

The Police Command's Side

The first result of this section establishes a necessary condition for the police's optimal response, given the equilibrium crime intensity, corresponding to condition (i) of the equilibrium.

Lemma 1. *Suppose that there is an equilibrium with $\lambda^* \in \mathcal{C}_{++}$ and that $C > \underline{C}(\mu^*)$. Then, the police activity at x is increasing in the crime intensity $\mu^*(x)$ and*

$$\forall x \in W, \quad z^*(x) = \ln \mu^*(x) + C - \frac{\int_W \ln \mu^*(x') dx'}{|W|}, \quad (3)$$

where $z^*(x) = \int_W K(\|x - y\|) \lambda^*(y) dy$ is the equilibrium police activity at x .

In fact, this lemma holds for any continuous crime intensity, not necessarily the equilibrium one. From Eq. (1), we already know that the police command tends to allocate more resources to regions where μ^* is high. Eq. (3) further specifies how: the optimal police activity at location x is an affine function of $\ln(\mu^*)$.

The intuition behind this result is straightforward. The probability of clearing a crime is a concave function of police activity due to inefficiencies arising from overlapping patrol coverage. As a result, the optimal allocation of patrols is achieved when the marginal utility gain—i.e., the expected number of cleared crimes—is equalized across all points in W .

The Criminals' Side

Let us turn to the criminal's side of the equilibrium. We show that potential criminals adopt a cutoff strategy to decide whether to commit a crime, and we derive necessary conditions for their location strategies and the total number of criminals—corresponding to conditions (ii) and (iii) of the equilibrium.

¹³Observe that the number of points in Φ can have any distribution and we assume that $\Pr(\Phi = \emptyset)$ is negligible, so this non-defined case has no impact on this average.

Eq. (2) implies that, if there is an equilibrium, a criminal with an opportunity cost κ commits a crime whenever

$$\max_{s \in \mathcal{S}} \left\{ \int_W \tilde{a}(x) e^{-z^*(x)} s(x) dx \right\} > \kappa + b.$$

The term $e^{-z^*(x)} \times \tilde{a}(x)$ is the product of the probability of committing a successful crime and the adjusted attractiveness at x . This term minus b is the expected payoff from committing a crime at x .

Since $\int_W \tilde{a}(x) e^{-z^*(x)} s(x) dx - b$ represents the mean of these benefits under the location strategy s (i.e., the distribution over W), any strategy with support in the set $\operatorname{argmax}_{x \in W} \{e^{-z^*(x)} \tilde{a}(x)\}$ can be the optimal response of a criminal to a patrolling intensity λ^* . This argmax set is non-empty, provided that $\tilde{a}$ and λ^* are continuous. Thus, if $\hat{x}$ is an element of this set, for any optimal location strategy,

$$\max_{s \in \mathcal{S}} \left\{ \int_W \tilde{a}(x) e^{-z^*(x)} s(x) dx \right\} = \tilde{a}(\hat{x}) e^{-z^*(\hat{x})}.$$

This implies that a criminal commits a crime if $\kappa < \tilde{a}(\hat{x}) e^{-z^*(\hat{x})} - b$. That is, a potential criminal commits a crime if and only if their opportunity cost is below a certain cutoff.

Moreover, because the cost of committing a crime is independent across individuals, the probability that a randomly chosen criminal commits a crime is $F_\kappa(\tilde{a}(\hat{x}) e^{-z^*(\hat{x})} - b)$. Recalling that the number of potential criminals is N , the expected number of crimes is $N F_\kappa(\tilde{a}(\hat{x}) e^{-z^*(\hat{x})} - b)$.

Characterization of the Equilibrium Intensities

We combine conditions (i)-(iii) of the equilibrium to characterize the equilibrium crime rate, crime and police intensities, as well as the ACR.

Proposition 1. *Suppose that $C > \underline{C}(\tilde{a})$. Then, in equilibrium, if $\lambda^* \in \mathcal{C}_{++}$, the police activity is given by*

$$\int_W K(\|x - y\|) \lambda^*(y) dy = z^*(x) = \ln f_{\tilde{a}}(x) + C + \mathcal{H}(u, f_{\tilde{a}}), \quad \forall x \in W, \quad (4)$$

the aggregate clearance rate in W is

$$p_W^* = 1 - e^{-C} \times e^{-D(u \| f_{\tilde{a}})}, \quad (5)$$

the value of crime is $V^* := (1 - p_W^*) \times \tilde{a}(W) - b$, the crime rate is $M^* := NF_\kappa(V^*)$, and the crime intensity is given by

$$\mu^*(x) = M^* \frac{\tilde{a}(x)}{\int_W \tilde{a}(x') dx'} = M^* f_{\tilde{a}}(x), \quad \forall x \in W. \quad (6)$$

This proposition establishes several interesting results. First, from Eq. (4), the police activity $z^*(x)$ is an increasing but concave function of $a(x)$: more attractive locations receive more police protection, but at a decreasing rate. The concavity reflects the spatial spillovers of police patrols: doubling police activity at a point (roughly, the number of patrols covering that point) does not double the clearance rate, for the same reason that doubling the number of coin tosses does not double the probability of obtaining heads at least once—each additional toss contributes less as success becomes increasingly likely. As a result, concentrating police protection in an already well-covered area yields diminishing gains in deterrence, and each increment in crime attractiveness triggers a smaller increment in police protection. In addition, Eq. (4) is the base to understand the spillovers of police protection, studied in detail in Section 5.

Second, the ACR, given by Eq. (5), depends positively on the police capacity C and on the relative entropy $D(u \| f_{\tilde{a}})$ between the uniform distribution and the distribution of adjusted attractiveness. The latter measures the spatial concentration of $\tilde{a}$: it equals zero when attractiveness is uniform across W , and increases as crime becomes more predictably concentrated. Crucially, concentration benefits the police: when $\tilde{a}$ is uniform, officers cannot do better than spreading resources evenly, whereas a more heterogeneous distribution allows police to exploit predictable crime patterns and achieve a strictly higher clearance rate. Beyond its closed-form simplicity, this functional form is the unique one giving discrete models a well-behaved PSPF (see Corollary 2 and the Supplementary Online Material for further details).

Third, although the police parameters influence the crime rate M^* , in equilibrium, they do not impact the spatial distribution of crime f_{μ^*} , which mirrors the distribution of adjusted attractiveness as shown by Eq. (6). This reflects a spatial no-arbitrage condition: in equilibrium, the police make criminals indifferent across locations, but the distribution of criminals must simultaneously make that police allocation optimal—and the distribution satisfying this condition is $f_{\tilde{a}}$.

Finally, we claim that M^* represents the minimum crime rate that any police deployment can achieve. The intuition follows from the no-arbitrage condition: z^* equalizes criminal payoffs across all locations, leaving no area relatively unprotected. Any real-

location of police resources must over-protect some locations and under-protect others, concentrating crime where payoffs are highest and raising the aggregate crime rate. This is formalized in the following corollary.

Corollary 1. *The police activity given by z^* is the only one minimizing the crime rate.*

4.3 Consistent Aggregability of the Continuous Model

The lack of consistent aggregation in the models of Merlo (2003) and Fu and Wolpin (2018) is common to all other discrete-choice models of crime in the literature. A natural question is therefore whether a model allowing for consistent aggregation exists. That is, is there a functional form linking clearance rates and police expenditures that allows consistent aggregation?

Our continuous model provides an answer to this question. Consider a discrete model induced by the PSPF given by Eq. (5). That is, consider an arbitrary partition of W , $\{W_i\}_{i \in I}$, and for each i , define the PSPF as if we applied our model within the cell, obtaining

$$\pi_{W_i}(\Lambda') = 1 - e^{-\frac{\Lambda'\theta}{|W_i|}} e^{-D_{W_i}(u \| f_{\tilde{a}})}, \quad (7)$$

for each $\Lambda' \geq 0$ and a positive constant θ , where D_{W_i} is the Kullback–Leibler divergence with domain in W_i , with $\tilde{a}$ normalized over W_i .

This PSPF is a one-parameter function, indexed by θ (the discrete-model counterpart of q), that depends on three aggregate quantities over W' : the area of W' , given by $\int_{W'} 1 \, dx$; the average attractiveness in W' , given by $\frac{1}{|W'|} \int_{W'} a(x) \, dx$; and a term related to the relative entropy of $a + b$ with respect to the uniform distribution, given by $\int_{W'} \ln(a(x) + b) \, dx$. In addition, by comparing with the ACR in the continuous case, we identify $\Lambda'\theta/|W'|$ as the police capacity in the corresponding discrete case.

Then, as we show in the following corollary, this discrete model satisfies the consistent aggregation principle.

Corollary 2. *A discrete model induced by the PSPF given by Eq. (7) respects the consistent aggregation principle.*

For brevity, we call a discrete model induced by the PSPF in Eq. (7) a *canonical discrete model*. These models are important for two reasons. First, their PSPF is microfounded. It arises endogenously from the topological model rather than being imposed by the modeler. Second, the family of PSPFs defining them is the unique

one respecting a set of desirable properties, which we formalize and characterize in the Supplementary Online Material.

To illustrate Corollary 2 numerically, we use the PSPF given by Eq. (7) in a discrete model and we set θ so as to yield an ACR of 0.7 under Split A. Table 2 shows that this discrete model respects the consistent aggregation principle.

Table 2: Aggregate Clearance Rate and Crime Rate by split (PSPF given by Eq. (7))

Clearance Technology	Split	# Units	ACR	Total crime
Canonical discrete model	Split A	2	0.7000	0.0152
	Split B	2	0.7000	0.0152
	Split C	4	0.7000	0.0152
	Split D	4	0.7000	0.0152

Notes: We set $\Lambda = 100$ (which induces interior solutions in all cases), $b = 0.1$, $N = 1$, F_κ as the uniform distribution on $[0, 1]$. We compute $\theta = 0.0096$, yielding the ACR of 0.7.

5 Spatial Spillovers and the Police Location

Proposition 1 characterizes the pointwise relation between police activity z^* and the attractiveness. Recall that the quantity $z^*(x)$ captures the average protection point x receives from the police (related to the number of units close to x and their respective distances), and it is intrinsically connected with the police intensity (roughly, the actual location of police units), represented by λ^* , through $z^*(x) = \int_W K(\|x - y\|)\lambda^*(y)dy$. In this section, we characterize how these spatial spillovers, embedded in the kernel K , shape the deployment of police units λ^* . Recall that $\mu^* = \text{const} \times (a + b)$ and $z^* = \ln(\text{const} \times (a + b))$, so z^* , a , and μ^* are related by strictly increasing transformations. Hence, characterizing how λ^* relates to any one of them suffices to understand its relation to all three.

We proceed in three steps: we first analyze the behavior of λ^* as $r \rightarrow 0$, establishing a connection with the discrete model; we then characterize λ^* for small r and its properties; finally we consider the general case.

All results hold for a two-dimensional city with an arbitrary crime-detecting kernel; the numerical illustrations use their one-dimensional, uniform-kernel counterparts for ease of exposition.

5.1 No Spillovers: Vanishing Coverage Radius

Our benchmark is the police allocation in a discrete model whose domain is partitioned into small regions. More specifically, consider a canonical discrete model. Let $\{W_i\}_{i=1}^n$ be a partition of W with $|W_i| = |W|/n$ and $\lim_{n \rightarrow \infty} \text{diam}(W_i) = 0$ for each i . Then, fixing $x_i \in W_i$ we define

$$\lambda_0^*(x_i) := \lim_{n \rightarrow \infty} \frac{\Lambda_{W_i}^*}{|W|/n}.$$

Proposition 2. *Consider a canonical discrete model with a fixed police capacity C . If the continuous model has the same police capacity, then $f_{\lambda_0^*} = f_{z^*}$, where z^* is given by Eq. (4). If, in addition, the continuous model has the same budget as in the canonical discrete model, then, as $r \rightarrow 0$, the police allocations in the discrete and continuous models coincide:*

$$\lim_{r \rightarrow 0} \frac{\lambda^*}{\lambda_0^*} \equiv 1.$$

This proposition establishes two results. First, the distribution of the police allocation in the canonical discrete model, as regions tend to singletons, coincides with the distribution of police activity in a continuous model with the same police capacity. Second, if the police budget also coincides, and we consider a continuous model without spillovers ($r \rightarrow 0$), the police allocations of the discrete and continuous models agree pointwise.

5.2 Local Spillovers: Second-Order Approximation

We now drop the assumption that $r \rightarrow 0$ and consider instead a coverage radius $r > 0$ small enough for the second-order Taylor approximation to be accurate. Our first result compares the police intensity when r is small to the police allocation of a canonical discrete model with the same police capacity and budget.

Proposition 3. *Fix a police capacity C and a budget Λ in both a canonical discrete model and a continuous model. Then, if r is small,*

$$\lambda^*(x) = \lambda_0^*(x) - \eta r^2 \nabla^2 \lambda_0^*(x) + O(r^4), \quad (8)$$

where η is a positive dimensionless constant depending on the shape of the crime-detecting kernel, but not on the coverage radius.

Spillovers distort the police allocation, and the stronger the spillover, the larger the distortion relative to the baseline.¹⁴ In particular, at local maxima of λ_0^* —where the Hessian is negative definite implying a negative Laplacian—the continuous allocation $\lambda^*(x)$ exceeds $\lambda_0^*(x)$.

To understand the economic intuition, start with a model without spillovers. At a local maximum of λ_0^* , x^M , (negative Laplacian), the surrounding areas have a lower police spatial density than at x^M . As we introduce spillovers with a positive r , a police unit deployed at x also protects the surrounding area $B(x, r)$. Keeping the discrete allocation λ_0^* would imply that the spillovers received at x^M from neighboring units—which have lower λ_0^* —are weaker than the spillovers sent out from x^M to its neighbors. To restore optimality, the police command must compensate by deploying more police at x^M , giving $\lambda^*(x^M) > \lambda_0^*(x^M)$. By the same logic, at a local minimum x^m (positive Laplacian), the spillovers received from neighbors are stronger than those sent out, reducing the need for local deployment, so $\lambda^*(x^m) < \lambda_0^*(x^m)$.

This proposition also establishes conditions ensuring a well-defined λ^* in Proposition 1. Combining Eq. (8), $\lambda_0^* = z^* \Lambda / (C|W|)$ (from the proof of Proposition 3), and $z^* > 0$, it suffices that $r^2 < \frac{1}{\eta} \min \left\{ \frac{z^*(x)}{\nabla^2 z^*(x)} : \nabla^2 z^*(x) > 0 \right\}$ to ensure $\lambda^* \in \mathcal{C}_{++}$. In this case, condition (i) in our equilibrium definition is met and the equilibrium exists.

In general, the interaction between the curvature of λ_0^* and the size of the spillovers governs both the direction and the magnitude of the distortion: the more curved the allocation at x and the larger the coverage radius, the stronger the imbalance between spillovers sent and received, and the larger the deviation of $\lambda^*(x)$ from $\lambda_0^*(x)$, as captured by the term $-\eta \nabla^2 \lambda_0^*(x) r^2$.

Numerical illustration of Proposition 3. Consider a one-dimensional city in $[-1, 1]$, so that the second-order approximation gives $\lambda^*(x) = \lambda_0^*(x) - \frac{r^2}{6} (\lambda_0^*)''(x) + O(r^4)$. Figure 6 shows this approximation for coverage radii 0.10 and 0.25 when $\lambda_0^* = 1 + x^2 - x^4$. Since $(\lambda_0^*)'' < 0$ at local maxima and $(\lambda_0^*)'' > 0$ at local minima, stronger spillovers raise the police intensity at the maxima and lower it at the minima. At inflection points, where $(\lambda_0^*)'' = 0$, the police intensity is unchanged to this order.

¹⁴We can write the second-order approximation relative to the crime-detecting kernel more generally as

$$\lambda^* = \lambda_0^* - \frac{\int_0^r K(\rho) \rho^3 d\rho}{4 \int_0^r K(\rho) \rho d\rho} \nabla^2 \lambda_0^* + O \left(\left(\frac{\int_0^r K(\rho) \rho^3 d\rho}{4 \int_0^r K(\rho) \rho d\rho} \right)^2 \right).$$

This expression makes explicit how λ^* depends on the shape of the kernel, independently of r .

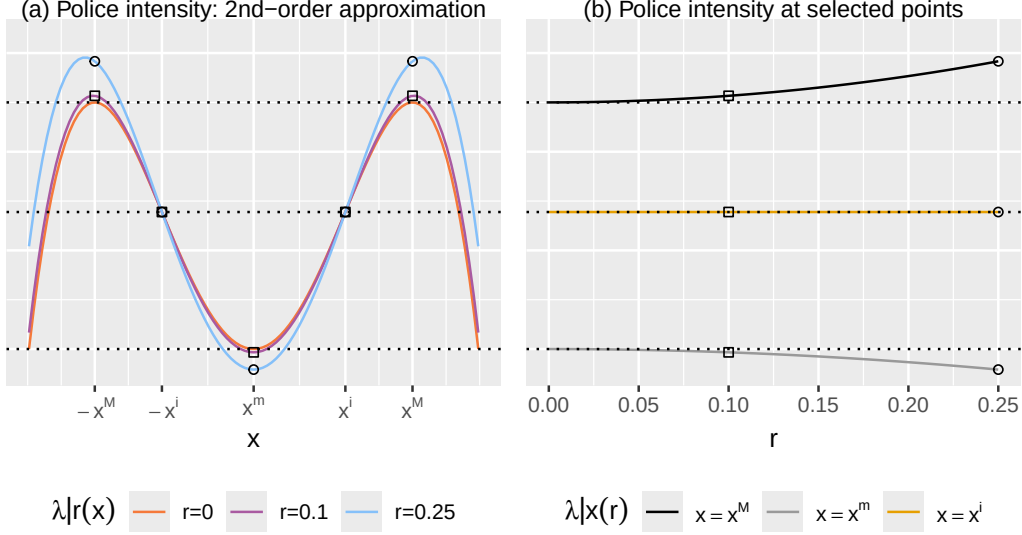


Figure 6: Panel (a) shows the second-order approximation of the police intensity for different radii ($r \rightarrow 0$, $r = 0.1$, and $r = 0.25$) in a one-dimensional city $W = [-1, 1]$ when $\lambda_0^*(x) = 1 + x^2 - x^4$. Panel (b) shows the change in the police intensity at the peak x^M , the valley x^m , and inflection point x^i of λ_0^* as r increases.

Peak Location Change

Also, from using the approximation given by Eq. (8), we can deduce the direction in which the local maxima and local minima change when spillovers are introduced.

Corollary 3. *Let x_0^c be a critical point of λ_0^* with non-singular Hessian $H_{\lambda_0^*}(x_0^c)$. Then the corresponding critical point of λ^* is located at $x^c = x_0^c + x_\epsilon + O(r^4)$, where*

$$x_\epsilon = \eta r^2 [H_{\lambda_0^*}(x_0^c)]^{-1} \nabla (\nabla^2 \lambda_0^*)(x_0^c).$$

This means that the maximal police intensity in the continuous model need not coincide with the crime hotspot identified by the discrete model. Because the Hessian is negative definite at local maxima, if x_0^c is a maximum, the shift x_ϵ is opposite to $\nabla(\nabla^2 \lambda_0^*)(x_0^c)$: the continuous model moves the police intensity toward the direction where the Laplacian is more negative, i.e., toward the sharper side of the peak. In other words, spatial spillovers pull police off-center toward the direction where the peak's curvature decreases fastest, because coverage from neighboring locations compensates less effectively on the sharper side of the peak. In particular, when λ_0^* is radially symmetric around x_0^c , we have $\nabla(\nabla^2 \lambda_0^*)(x_0^c) = 0$ and the peak of λ^* remains at x_0^c .

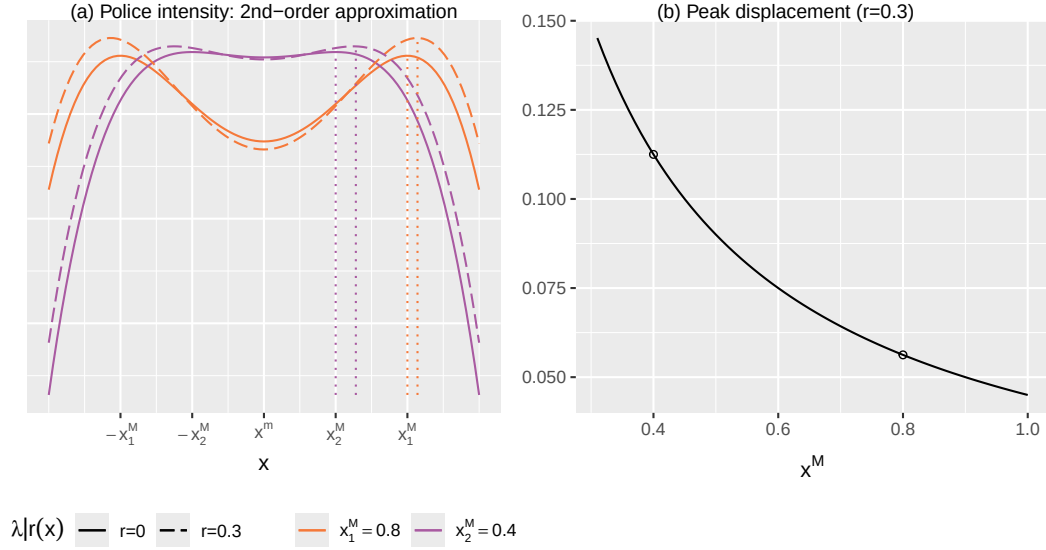


Figure 7: Panel (a) shows the second-order approximation of the police intensity for distinct configurations of crime attractiveness (differing by the distance between peaks) and two coverage radii ($r \rightarrow 0$ and $r = 0.3$) when $W = [-1.2, 1.2]$. We used $\lambda_0^*(x) = a_0(p) + 2p^2x^2 - x^4$, where p is the position of the peak and a_0 is used to normalize λ_0^* . Panel (b) shows the peak displacement for several peak distances when $r = 0.3$.

Numerical illustration of Corollary 3. Consider one-dimensional cities in $[-1.2, 1.2]$, so $x_\epsilon = \eta r^2 (\lambda_0^*)'''(x_0^c) / (\lambda_0^*)''(x_0^c)$. Let $\lambda_0^*(x)$ be a quartic polynomial. Panel (a) of Figure 7 depicts λ_0^* for two cities with distinct peak distances and the corresponding peaks of the police intensity when $r = 0.3$. When peaks are far apart, they are more symmetric and the displacement x_ϵ is small. When peaks are close together, one side of each peak is steeper, generating a larger displacement. Panel (b) of Figure 7 shows the peak displacement as a function of the right-side peak's position in λ_0^* , for $r = 0.3$. This illustrates how the spatial structure of λ_0^* —not just its value at the peak—is crucial to determine the police allocation in our continuous model.

Entropy Change

Proposition 3 is a local result: at each x , the continuous allocation $\lambda^*(x)$ deviates from the discrete benchmark $\lambda_0^*(x)$ in a direction that amplifies peaks and valleys, suggesting that λ^* is less dispersed than λ_0^* at the aggregate level. Proposition 4 confirms this intuition by comparing the two functions at the aggregate level.

Proposition 4. Suppose that $\nabla a(x) \cdot \mathbf{n}(x) = 0$ for $x \in \text{Bd}(W)$, where $\mathbf{n}(x)$ is the

outward unit normal to $\text{Bd}(W)$ at x . Then, if the canonical discrete model and the continuous model have the same budget and police capacity:

$$\mathcal{H}(f_{\lambda^*}) = \mathcal{H}(f_{\lambda_0^*}) - \eta r^2 \mathcal{I}(f_{\lambda_0^*}) + O(r^4),$$

where $\mathcal{I}(f_{\lambda_0^*})$ is the Fisher information of $f_{\lambda_0^*}$, which is non-negative. In particular, for r small enough, $\mathcal{H}(f_{\lambda^*}) < \mathcal{H}(f_{\lambda_0^*})$ whenever λ_0^* is non-constant, with the entropy loss proportional to the Fisher information of the discrete benchmark.

The boundary condition $\nabla a(x) \cdot \mathbf{n}(x) = 0$ on $\text{Bd}(W)$ simply states that W is surrounded by areas with the same attractiveness as its boundary, so that boundary effects do not distort the interior allocation.

Proposition 4 confirms that the amplification of peaks and valleys translates into a police intensity with lower entropy when spillovers are present, but also measures this difference. Intuitively, a more spatially heterogeneous baseline allocation λ_0^* —reflected in a higher Fisher information—exhibits more pronounced curvatures, providing more scope for spillovers to further amplify these differences and thus generating a larger entropy loss. This effect interacts with the size of the spillovers: a larger coverage radius r means that each police unit protects a wider area, strengthening the spillovers and amplifying the distortion. It is therefore the product $\eta r^2 \mathcal{I}(f_{\lambda_0^*})$ that governs the entropy loss—the spatial heterogeneity of the baseline and the strength of the spillovers reinforce each other.

Numerical illustration of Proposition 4. Let $W = [-\pi, \pi]$ and $\lambda_0^*(x) = \cos(nx) + 2$, where n is a positive integer. This function ensures that $\nabla a(x) \cdot \mathbf{n} = (\lambda_0^*)'(x) = 0$ for $x \in \text{Bd}(W)$, since $\sin(n \cdot (\pm\pi)) = 0$ for integer n . The entropy of $f_{\lambda_0^*} = (\cos(nx) + 2)/(4\pi)$ does not depend on n (each period is an identical copy of the first, so faster oscillations add no new uncertainty), while the Fisher information increases linearly with n^2 (faster oscillations produce steeper slopes). Thus, as $r \rightarrow 0$, the entropy of f_{λ^*} converges to that of $f_{\lambda_0^*}$, which is constant in n . For small $r > 0$, however, the entropy decreases proportionally to $(nr)^2$.

Taken together, Proposition 3, Corollary 3 and Proposition 4 establish that spillovers are not merely a technical feature of the continuous model but have potentially substantive consequences for the police allocation through equilibrium effects that discrete models fail to capture.

5.3 Global Spillovers: Higher-Order Approximations

The results from Subsection 5.2 rely on a second-order approximation of λ^* , with remainder $O(r^4)$. However, some phenomena generated by spillovers can only be captured with higher-order approximations. To illustrate this limitation, we discuss peak reversal, whereby a local maximum of attractiveness becomes a local minimum of the police intensity, thereby generating new local maxima. This can occur when police units placed at the peak already cover the entire attractive region through spillovers and additional units at the same location overlap heavily and contribute little marginal deterrence. It becomes more efficient to flank the peak—placing units on either side so that their coverage areas jointly protect the peak while also reaching areas beyond it.

However, this redistribution of enforcement involves spatial scales beyond the immediate neighborhood of the peak, suggesting that a second-order approximation is too coarse to detect it. This is shown formally in the following claim.

Claim 1. *The second-order approximation cannot certify a peak reversal: at the radius where reversal first appears, the retained correction is as large as the leading term, so the approximation's error is no longer negligible.*

Peak Reversal is Possible under a Fourth-Order Approximation

We show that a higher-order approximation can generate a peak reversal. Our starting point is the fourth-order approximation of $\lambda^*(x)$, given by:

$$\lambda^*(x) = \lambda_0^*(x) - \eta r^2 \nabla^2 \lambda_0^*(x) + \tilde{\eta} r^4 \nabla^4 \lambda_0^*(x) + O(r^6),$$

where the coefficients η and $\tilde{\eta}$ depend on the shape of the kernel but not on the coverage radius.¹⁵ Then, we use this expansion to establish the following claim.

Claim 2. *Let $\lambda_0^*(x_1, x_2) = a_4(x_1^2 + x_2^2)^2 + a_2(x_1^2 + x_2^2) + a_0$ with $a_2, a_4 < 0$, so the origin is the only peak of λ_0^* and let $\bar{r} := \sqrt{a_2/(16\eta a_4)}$. If $r > \bar{r}$, then peak reversal occurs at the origin and $\lambda^*(0)$ becomes a local minimum. Moreover, the local maxima form a circle of radius $\sqrt{8\eta(r^2 - \bar{r}^2)}$.*

This result implies that, beyond the peak-reversal threshold radius $\bar{r}$, the radius of the circle of maxima increases with r : larger spillovers push the optimal enforcement zones farther apart.

¹⁵Let $q^{(n)} := 2\pi \int_0^1 \bar{K}(\rho) \rho^n d\rho$, then $\eta = q^{(3)}/(4q^{(1)})$ and $\tilde{\eta} = (4(q^{(3)})^2 - q^{(1)}q^{(5)})/(64(q^{(1)})^2)$.

Note that, although the result uses the exact formula for a fourth-degree polynomial rather than a small- r approximation, it does not require r to be large relative to the city: even a small coverage radius can generate peak reversal provided $r > \bar{r}$, which can be made arbitrarily small by choosing a_2 close to zero.

Interestingly, for this polynomial the second-order approximation yields the same threshold $\bar{r}$ and radius of the local maxima as the exact formula. However, as predicted by Claim 1, the estimation of λ^* would contain a large error.

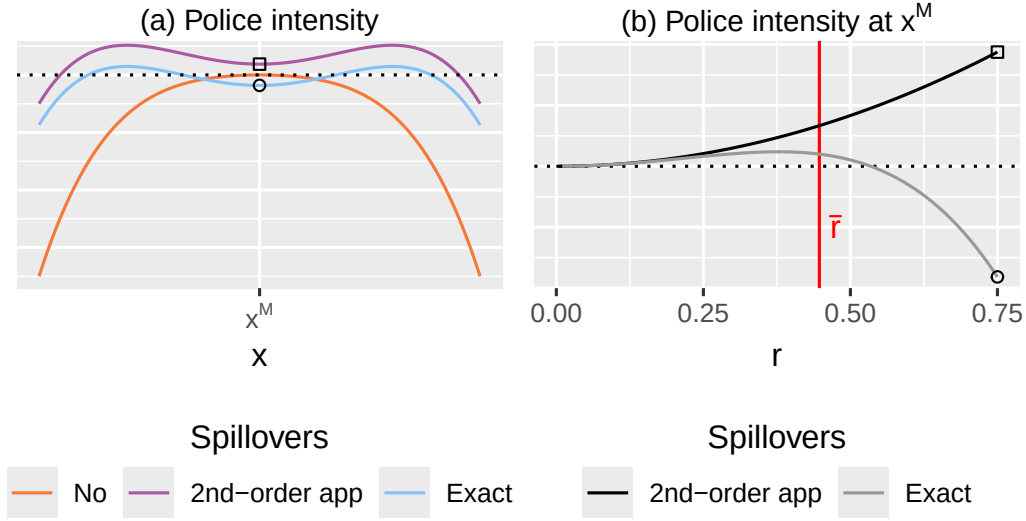


Figure 8: The criminal attractiveness generates $\lambda_0^*(x) = 4 - x^2 - \frac{5}{2}x^4$ when $W = [-1, 1]$. Panel (a) shows λ_0^* (when there are no spillovers), the second-order approximation and the exact value of λ^* for $r = 0.75$. Panel (b) depicts how the police intensity changes with r at the peak for the second-order approximation and the exact value of $\lambda^*(x^M)$, as well as the threshold $\bar{r}$.

Numerical illustration of Claims 1 and 2. Consider a city in $[-1, 1]$ in which $\lambda_0^*(x) = a_4x^4 + a_2x^2 + 1$ with $a_4, a_2 < 0$. Since λ_0^* is a fourth-degree polynomial, the fourth-order approximation of λ^* corresponds to the exact police intensity. In this case, $\bar{r} = \sqrt{a_2/(2a_4)}$ and the local maxima of λ^* are located at $\pm\sqrt{r^2 - \bar{r}^2}$. Panel (a) of Figure 8 shows λ_0^* , λ^* , and the local maxima of λ^* . Panel (b) shows how $\lambda^*(0)$ varies with r and compares it with the second-order approximation of λ^* at zero. Observe that, until approximately $\bar{r}/2$, the second-order approximation introduces only a small error, but at $r = \bar{r}$, when peak reversal starts to occur, the second-order approximation is no longer valid.

Beyond the Fourth-Order Approximation

The analysis above is primarily intended to illustrate that, depending on the context, higher-order approximations can introduce more qualitative richness in the analysis than the second-order approximation, rather than to describe a practically relevant case. For the uniform crime-detecting kernel, there is a closed-form infinite series to represent λ^* while, for a non-uniform kernel, the Taylor expansion of λ^* in powers of r can be computed term by term to any desired order, with each coefficient determined by the corresponding moment of the kernel.¹⁶

6 Conclusion

In this paper, we show the advantages of using a topological rather than a discrete-choice model of crime to study spatial patterns of crime and police deployment. We begin by showing that existing spatial models of crime suffer from a failure of internal consistency: equilibrium outcomes depend on how the city is partitioned. We further argue that current frameworks cannot determine the equilibrium police allocation. Their reduced-form structure only yields aggregate quantities when cells are large, while the absence of an explicit topological structure precludes the incorporation of spillovers, which is particularly problematic when cells are small.

To address these limitations, we propose a model drawing on tools from the theory of point processes, endowed with an explicit topological structure, and with assumptions grounded in recent empirical evidence. It solves the issue of consistent aggregation since the PSPF it endogenously generates, when used within each cell of a discrete model, renders equilibrium outcomes invariant to the partition.

In addition, it allows us to determine the equilibrium police allocation and to measure the bias a discrete model introduces when ignoring spillovers of police protection. We find that, when the coverage radius is small, a model without spillovers underestimates (overestimates) police intensity at local maxima (minima) of crime attractiveness. As a result, the police allocation becomes more concentrated, and this is reflected in a lower entropy of the police allocation. Moreover, while a model without spillovers predicts peaks of police spatial density at maxima of attractiveness, as spillovers are introduced, these peaks shift toward the steepest side (more precisely, the sharpest cur-

¹⁶For the uniform kernel, we can use properties of the Fourier transform to obtain $\lambda_0^*(x) = \sum_{k=0}^{\infty} \frac{r^{2k}}{4^k k! (k+1)!} \nabla^{2k} \lambda^*(x)$. Inverting this series term by term recovers the approximation coefficients.

vature side) of the attractiveness function. We also illustrate a qualitative result that cannot be obtained without a topological structure: a local maximum of crime attractiveness can become a local minimum of police intensity when the coverage radius is above a threshold. These results underscore the importance of taking the spatial dimension into account whenever police units offer protection to nearby locations. Ignoring spatial spillovers may introduce significant biases.

Our model is also tractable, enabling a clearer understanding of the forces that drive spatial patterns of crime and police deployment and allowing us to derive policy implications. Furthermore, it is flexible: while we focus on attractiveness as the sole source of spatial heterogeneity—since our main interest is spillovers of police protection—the model remains tractable if we introduce heterogeneity along other dimensions, such as the reward for clearing crimes, the punishment for crimes, or the clearance technology. We leave these extensions for future work.

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Appendices

A Equilibrium of the Discrete Model

We solve the equilibrium assuming that the equilibrium expenditures on police are strictly positive for each W_i . One way to interpret this assumption is that we focus only on the regions of the jurisdiction that are attractive enough to induce crime and, therefore, require enforcement. Alternatively, this assumption can be seen as a consequence of having a sufficiently high budget Λ . Finally, we avoid excessive formalism, as solving this model does not yield insights beyond those already present in the literature.

We start with the criminals. The average profit of a successful crime in region W_i is $a(W_i)$, thus, the expected profit from committing a crime in region W_i in equilibrium is

$$a(W_i)(1 - \pi_{W_i}(\Lambda_i^*)) - b \pi_{W_i}(\Lambda_i^*).$$

In equilibrium, criminals must be indifferent as to which region they commit a crime in. This is a well-known result: if there were a region W_i preferred over all others, then all crime would be concentrated there. This would prompt the police command to allocate all the resources to that region. Facing high police enforcement, W_i would no longer yield high expected profits for criminals, contradicting the assumption that W_i is the preferred region.

Therefore, the equilibrium value of committing a crime in each region is constant, which we denote as V^* . Note that this value might, in principle, depend on the partition of W and on the functionals $(\pi_{W_i})_{i \in I}$. A necessary condition for equilibrium is then

$$(a(W_i) + b)(1 - \pi_{W_i}(\Lambda_{W_i}^*)) = V^* + b.$$

By inverting the π_{W_i} functions, we can determine the optimal equilibrium expenditure levels for each region, given by

$$\Lambda_{W_i}^* = \pi_{W_i}^{-1}(1 - (V^* + b)/(a(W_i) + b)). \quad (9)$$

Now, we can analyze the problem of the police command. Recall that M_i^* is the equilibrium number of crimes in region W_i . The police maximize the total expected

apprehensions, subject to the budget constraint, which is given by

$$\max_{\Lambda_1, \Lambda_2, \dots, \Lambda_n} \sum_{i \in I} M_i^* \pi_{W_i}(\Lambda_i) \quad \text{s.t.} \quad \sum_{i \in I} \Lambda_i \leq \Lambda$$

If we attach the Lagrangian multiplier ζ to the budget constraint, the first-order condition of the problem implies that

$$M_i^* = \zeta / \pi'_i(\Lambda_{W_i}^*) \quad (10)$$

whenever $M_i^* > 0$. Since we assume that there is police enforcement in each region in equilibrium, $M_i^* > 0$ for each $i \in I$.

Thus, we can write the ACR using the partition $\{W_i\}_{i \in I}$ as

$$p_{\{W_i\}_{i \in I}}^* = \frac{\sum_{i \in I} M_i^* \pi_i(\Lambda_i^*)}{\sum_{i \in I} M_i^*}.$$

Using Eqs. (9) and (10), we can write the ACR down as a function of the fundamentals and V^* :

$$p_{\{W_i\}_{i \in I}}^* = 1 - (V^* + b) \frac{\sum_{i \in I} \frac{\pi_{W_i}^{-1'}(1 - (V^* + b)/(a(W_i) + b))}{a(W_i) + b}}{\sum_{i \in I} \pi_{W_i}^{-1'}(1 - (V^* + b)/(a(W_i) + b))}.$$

Finally, V^* is obtained using the budget constraint:

$$\sum_{i \in I} \pi_{W_i}^{-1}(1 - (V^* + b)/(a(W_i) + b)) = \sum_{i \in I} \Lambda_{W_i}^* = \Lambda.$$

The model is, thus, completely characterized.

B Proofs

Proof of Eq. (1)

Using the independence between Φ and Ψ , we have that

$$U^{\text{pol}}(\lambda, \mu) = \mathbb{E} \left[\sum_{x \in \Phi} \left(1 - \mathbb{E} \left[\prod_{y \in \Psi} (1 - K(\|x - y\|)) \mid \Phi \right] \right) \right].$$

Then, using the generating functional property, we have

$$\mathbb{E}\left[\prod_{y \in \Psi} (1 - K(\|x - y\|)) \mid \Phi\right] = e^{-\int_W K(\|x - y\|) \lambda(y) dy}$$

and thus

$$U^{\text{pol}}(\lambda, \mu) = \mathbb{E}\left[\sum_{x \in \Phi} \left(1 - e^{-\int_W K(\|x - y\|) \lambda(y) dy}\right)\right].$$

Finally, we use the intensity property in the above expression to obtain the expression of $U^{\text{pol}}(\lambda, \mu)$ as a function of μ and λ . $\square$

Proof of Eq. (2)

Since X and Ψ are independent, the payoff from committing a crime at X can be written as

$$\mathbb{E}\left[a(X) \mathbb{E}\left[\prod_{y \in \Psi} (1 - K(\|X - y\|)) \mid X\right] - b\left(1 - \mathbb{E}\left[\prod_{y \in \Psi} (1 - K(\|X - y\|)) \mid X\right]\right)\right].$$

Using the generating functional property, the above expression becomes

$$\mathbb{E}\left[a(X) e^{-\int_W K(\|X - y\|) \lambda(y) dy} - b\left(1 - e^{-\int_W K(\|X - y\|) \lambda(y) dy}\right)\right].$$

Taking the expectation and rearranging terms results in the desired criminal's utility expression. $\square$

Proof of Lemma 1

Proof. Observe that $C > \underline{C}(\mu^*)$ implies that $z^* > 0$ for each $x \in W$ and the budget constraint is satisfied, so the police activity described by Eq. (3) is feasible. The police command's problem can be equivalently formulated as minimizing $\int_W \exp(-\int_W K(\|x - y\|) \tilde{\lambda}(y) dy) \mu^*(x) dx$ with respect to $\tilde{\lambda}$ positive, continuous in W , and such that $\int_W \tilde{\lambda}(x) dx = \Lambda$. Take an arbitrary λ in this set. By Fubini's theorem and the symmetry of K , we have

$$\int_W \int_W K(\|x - y\|) \lambda(y) dy dx = \int_W \lambda(y) \int_W K(\|x - y\|) dx dy = \int_W \lambda(y) q dy = \Lambda q = C|W|.$$

Using Jensen's inequality and the convexity of $z \mapsto \exp(-z)$, we have

$$\begin{aligned}
& \int_W \exp\left(-\int_W K(\|x-y\|)\lambda(y)dy\right)\mu^*(x)dx \\
&= |W| \left(\frac{1}{|W|} \int_W \exp\left(-\int_W K(\|x-y\|)\lambda(y)dy + \ln(\mu^*(x))\right) dx \right) \\
&\geq |W| \exp\left(-\frac{1}{|W|} \int_W \left(\int_W K(\|x-y\|)\lambda(y)dy - \ln(\mu^*(x))\right) dx \right) \\
&= |W| \exp\left(-\frac{1}{|W|} \int_W \int_W K(\|x-y\|)\lambda(y)dy dx + \frac{1}{|W|} \int_W \ln(\mu^*(x)) dx \right) \\
&= |W| \exp\left(-C + \frac{1}{|W|} \int_W \ln(\mu^*(x)) dx \right).
\end{aligned}$$

It is easy to check that any λ satisfying (3) attains the bound above. In addition, any λ that does not satisfy (3) does not attain the bound, proving that measures that satisfy (3) are the only ones attaining the minimum. $\square$

Proof of Proposition 1

Proof. Observe that $C > \underline{C}(\tilde{a})$ implies that, for each $x \in W$, $z^*(x)$ given by (4) is positive. Moreover, the budget constraint is satisfied, so the police activity described by Eq. (4) is feasible. We first prove Eq. (4), followed by Eq. (6), and finally Eq. (5).

Police activity. We show that z^* defined by Eq. (4) is the only possible equilibrium police activity. In this part of the proof, μ^* is the equilibrium crime intensity, but we do not impose that it satisfies Eq. (6). Suppose, by way of contradiction, that there is an equilibrium police activity $\hat{z}$ given by an intensity measure $\hat{\lambda}$ that is different from λ^* . Observe that, for each $x \in W$

$$\tilde{a}(x)e^{-z^*(x)} = e^{-C}e^{-D(u\|f_{\tilde{a}})}\frac{1}{|W|}\int_W \tilde{a}(x')dx' := c^*.$$

That is, $\tilde{a}e^{-z^*}$ is constant. Since $\hat{z} \neq z^*$, $\tilde{a}e^{-\hat{z}}$ is not constant. Also, recall that we restrict $\hat{\lambda}$ to be continuous, so $\hat{z}$ is continuous and so is $\tilde{a}(x)e^{-\hat{z}(x)}$. Therefore, $\tilde{a}e^{-\hat{z}}$ attains a maximum in W , which we denote as $\hat{c}$. Because $\tilde{a}e^{-\hat{z}}$ is not constant, $\hat{A} := \{x \in W : e^{-\hat{z}(x)}\tilde{a}(x) = \hat{c}\}$ is a proper subset of W . Due to the maximizing behavior of criminals imposed by condition (ii) of the equilibrium, all crime must occur in $\hat{A}$ (i.e.,

we would need $\mu^*(x) = 0$ if $x \notin \hat{A}$, leading to

$$U^{\text{pol}}(\hat{\lambda}, \mu^*) = \int_W (1 - \hat{c}/\tilde{a}(x))\mu^*(x)dx.$$

We prove now that setting z^* would give the police command a greater payoff. Because $\hat{z}$ and z^* are distinct but both represent feasible police activities, we have $\int_W z^*(x)dx = \int_W \hat{z}(x)dx$. Hence, there exist points in W where $z^*(x) > \hat{z}(x)$, and others where $z^*(x) < \hat{z}(x)$. This means that the set $B := \{x \in W : \hat{z}(x) < z^*(x)\}$ is non-empty. In addition, observe that $\tilde{a}(x)e^{-z^*(x)} < \tilde{a}(x)e^{-\hat{z}(x)}$ if, and only if, $\hat{z}(x) < z^*(x)$, thus, $B = \{x \in W : \tilde{a}(x)e^{-z^*(x)} < \tilde{a}(x)e^{-\hat{z}(x)}\}$. Recalling that $\tilde{a}e^{-z^*} = c^*$, we have $B = \{x \in W : c^* < \tilde{a}(x)e^{-\hat{z}(x)}\}$. Let $x' \in \hat{A}$. Then, for any $x \in B$,

$$\hat{c} = \tilde{a}(x')e^{-\hat{z}(x')} \geq \tilde{a}(x)e^{-\hat{z}(x)} > c^* = \tilde{a}(x)e^{-z^*(x)},$$

implying that $\hat{A} \subset B$. In particular, if $x' \in \hat{A}$, then $(1 - e^{-z^*(x')}) > 1 - \hat{c}/\tilde{a}(x')$, yielding

$$\begin{aligned} U^{\text{pol}}(\hat{\lambda}, \mu^*) &= \int_W (1 - \hat{c}/\tilde{a}(x))\mu^*(x)dx = \int_{\hat{A}} (1 - \hat{c}/\tilde{a}(x))\mu^*(x)dx \\ &< \int_{\hat{A}} (1 - e^{-z^*(x)})\mu^*(x)dx = \int_W (1 - e^{-z^*(x)})\mu^*(x)dx = U^{\text{pol}}(\lambda^*, \mu^*), \end{aligned}$$

so $\hat{\lambda}$ does not maximize the police command's payoff, and condition (i) of the equilibrium is not satisfied. Thus, no police activity different from z^* can be the equilibrium intensity.

Crime intensity. First, we find the value of committing crime V^* . To do that, it suffices to replace z^* defined by Eq. (4) in the value of committing crimes, given by $\tilde{a}(x)e^{-z^*(x)}$ for any $x \in W$.

Now, we show that μ^* defined by Eq. (6) is the only possible equilibrium crime intensity. The police activity defined by Eq. (4) must be the best response for the crime intensity so that condition (i) of the equilibrium is satisfied. But the best response for the crime intensity is characterized by Eq. (3). Hence, we must have

$$\forall x \in W, \quad \ln \mu^*(x) + C - \frac{\int_W \ln \mu^*(x')dx'}{|W|} = \ln f_{\tilde{a}}(x) + C + \mathcal{H}(u, f_{\tilde{a}}).$$

Rewriting this relation, if there is μ^* solving this functional equation, it satisfies

$$\forall x \in W, \quad \ln \left(\frac{\mu^*(x)}{\tilde{a}(x)} \right) = \ln(c)$$

for a positive constant c , implying that $\mu^* = c \times \tilde{a}$. However, due to condition (iii) of equilibrium, $\int_W \mu^*(x) dx = \int_W c \tilde{a}(x) dx = M^*$, so the only value of c our equilibrium admits is $c = M^*/(\int_W \tilde{a}(x) dx)$.

ACR. Because Φ and Ψ are independent, and using the generating functional of a PPP, we have that

$$\begin{aligned} p_W^* &= \mathbb{E} \left[\frac{\sum_{x \in \Phi} \left(1 - \prod_{y \in \Psi} (1 - K(\|x - y\|)) \right)}{\sum_{x \in \Phi} \mathbb{1}_{\{x \in W\}}} \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[\frac{\sum_{x \in \Phi} \left(1 - \prod_{y \in \Psi} (1 - K(\|x - y\|)) \right)}{\sum_{x \in \Phi} \mathbb{1}_{\{x \in W\}}} \middle| \Phi \right] \right] \\ &= \mathbb{E} \left[\frac{\sum_{x \in \Phi} (1 - e^{-z^*(x)})}{\sum_{x \in \Phi} \mathbb{1}_{\{x \in W\}}} \right] \\ &= \sum_{m=0}^{\infty} \Pr \left(\sum_{x \in \Phi} \mathbb{1}_{\{x \in W\}} = m \right) \mathbb{E} \left[\frac{\sum_{x \in \Phi} (1 - e^{-z^*(x)})}{\sum_{x \in \Phi} \mathbb{1}_{\{x \in W\}}} \middle| \sum_{x \in \Phi} \mathbb{1}_{\{x \in W\}} = m \right] \end{aligned}$$

If $\{x_1, x_2, \dots, x_m\}$ represents Φ when the number of points is m , the expectation above can be written as

$$\mathbb{E} \left[\frac{\sum_{j=1}^m (1 - e^{-z^*(x_j)})}{m} \right] = \frac{1}{m} \sum_{j=1}^m \mathbb{E} [(1 - e^{-z^*(x_j)})] = \mathbb{E} [(1 - e^{-z^*(x)})].$$

Since the average distribution of a given crime is μ^*/M^* ,

$$\begin{aligned} p_W^* &= \sum_{m=0}^{\infty} \Pr \left(\sum_{x \in \Phi} \mathbb{1}_{\{x \in W\}} = m \right) \mathbb{E} [(1 - e^{-z^*(x)})] \\ &= \mathbb{E} [(1 - e^{-z^*(x)})] \\ &= \int_W (1 - e^{-z^*(x)}) \frac{\mu^*(x)}{M^*} dx \\ &= 1 - \int_W e^{-z^*(x)} f_{\tilde{a}}(x) dx. \end{aligned}$$

To conclude the proof, we replace the value of z^* given by Eq. (4) in the above expression and rearrange it to obtain Eq. (5). $\square$

Proof of Corollary 1

Proof. Consider the notation used to obtain the equilibrium police activity z^* in the proof of Proposition 1. Under the arbitrary police activity $\hat{z} \neq z^*$, all crime is committed in $\hat{A} \subset B$. Defining $\hat{V}$ as the payoff of criminals for committing a crime in B , we have

$$\hat{V} = \tilde{a}(x)e^{-\hat{z}(x)} - b > c^* - b = V^*,$$

for $x \in \hat{A}$. Therefore, the number of criminals under $\hat{z}$ is given by $NF_\kappa(\hat{V}) > NF_\kappa(V^*) = M^*$. $\square$

Proof of Corollary 2

Proof. Substituting Eq. (7) into the indifference condition and isolating the optimal police expenditures gives $\sum_{i \in I} \Lambda_{W_i}^* = \Lambda_{\cup_{i \in I} W_i}^*$, so condition (iii) for consistent aggregation is satisfied. In particular, we have

$$\Lambda = \frac{|W|}{\theta} [\ln(a(W) + b) - D(u \| f_{\tilde{a}}) - \ln(V^* + b)],$$

so V^* is partition-independent and so is $NF_\kappa(V^*)$, the aggregate crime rate. Therefore, (i) is satisfied. Finally, the indifference condition gives $1 - \pi_{W_i}(\Lambda_{W_i}^*) = (V^* + b)/(a(W_i) + b)$, so the ACR for both partitions $\{W_i\}_{i \in I}$ and $\{\cup_{i \in I} W_i\}$ reduces to $1 - [(V^* + b)|\cup_{i \in I} W_i|]/[\int_{\cup_{i \in I} W_i} (a(x) + b) dx]$, establishing condition (ii). $\square$

Proof of Proposition 2

Proof. Let Λ be the police budget. From Appendix A, in a discrete equilibrium model we have

$$V^* + b = (a(W_i) + b)(1 - \pi_{W_i}(\Lambda_{W_i}^*)).$$

for each $i \in I$. Since we use a consistent model, π_{W_i} is given by Eq. (7):

$$V^* + b = e^{-\frac{\Lambda_{W_i}^* \theta}{|W|/n}} e^{\frac{1}{|W|/n}} \int_{W_i} \ln \tilde{a}(x) dx.$$

Taking the log in both sides and summing across all sets in the partition gives

$$\begin{aligned}\sum_{i \in I} \ln(V^* + b) &= -\frac{\theta \sum_{i \in I} \Lambda_{W_i}^*}{|W|/n} + \frac{1}{|W|/n} \sum_{i \in I} \int_{W_i} \ln \tilde{a}(x) dx \\ &= -\frac{\Lambda \theta}{|W|/n} + \frac{1}{|W|/n} \int_W \ln \tilde{a}(x) dx,\end{aligned}$$

thus

$$\ln(V^* + b) = -\frac{\Lambda \theta}{|W|} + \frac{1}{|W|} \int_W \ln \tilde{a}(x) dx$$

and

$$\frac{\Lambda_{W_i}^* \theta}{|W|/n} = \frac{1}{|W|/n} \int_{W_i} \ln f_{\tilde{a}}(x) dx + \frac{\Lambda \theta}{|W|} + \mathcal{H}(u, f_{\tilde{a}})$$

for each $i \in I$. As $n \rightarrow \infty$, $1/(|W|/n) \int_{W_i} \ln f_{\tilde{a}}(x) dx \rightarrow \ln f_{\tilde{a}}(x_i)$ and we have

$$\theta \lambda_0^*(x) = \ln f_{\tilde{a}}(x) + C + \mathcal{H}(u, f_{\tilde{a}}), \quad \forall x \in W,$$

which coincides with $z^*(x)$ of a continuous model when the police capacity is C . Since λ_0^* is proportional to z^* , we have $f_{\lambda_0^*} = f_{z^*}$.

Consider now a continuous model with the same police capacity and budget Λ . Define K_1 as a crime-detecting kernel with coverage radius one and set $K_r(\rho r) := K_1(\rho)$, so that all kernels in the family share the same shape and differ only in their reach. Since $C = \theta \Lambda / |W|$ in the discrete model and the budget in the continuous model is the same, we have $C = q \Lambda / |W|$, thus $q = \int_W K_r(\|x - y\|) dy = \theta$ for every r .

Note that

$$\lambda_0^*(x) = \int_W \frac{K_r(\|x - y\|)}{\theta} \lambda^*(y) dy$$

for each r . Fix any continuous test function h on W . Multiplying the identity above by $h(x)$, integrating over x , and exchanging the order of integration,

$$\int_W h(x) \lambda_0^*(x) dx = \int_W \lambda^*(y) \left[\int_W \frac{K_r(\|x - y\|)}{\theta} h(x) dx \right] dy = \int_W \lambda^*(y) h_r(y) dy,$$

where

$$h_r(y) := \int_W \frac{K_r(\|x - y\|)}{\theta} h(x) dx.$$

Because h is uniformly continuous (being continuous on a compact set), the value of $\omega_r := \sup\{|h(x) - h(y)| : \|x - y\| < r\}$ is well-defined and goes to zero as $r \rightarrow 0$. Since

K_r/θ is nonnegative, integrates to one, and is supported on $B(y, r)$,

$$|h_r(y) - h(y)| = \left| \int_W \frac{K_r(\|x - y\|)}{\theta} (h(x) - h(y)) dx \right| \leq \int_W \frac{K_r(\|x - y\|)}{\theta} \omega_r dx \leq \omega_r$$

and

$$\left| \int_W h \lambda_0^* - \int_W h \lambda^* \right| = \left| \int_W (h_r - h) \lambda^* \right| \leq \omega_r \int_W \lambda^* \xrightarrow{r \rightarrow 0} 0.$$

Since this holds for every continuous h , λ^* converges weakly to λ_0^* as $r \rightarrow 0$. $\square$

Proof of Proposition 3

Proof. Let $\bar{K}(\rho) := K(r\rho)$ and $q^{(n)} := 2\pi \int_0^1 \bar{K}(\rho) \rho^n d\rho$. We represent the vectors x and y , respectively, by $[x_1 \ x_2]^T$ and $[y_1 \ y_2]^T$ and we let H be the Hessian operator. The Taylor expansion of $\lambda^*(y)$ around x is given by

$$\lambda^*(y) = \lambda^*(x) + \nabla \lambda^*(x)^T (y - x) + \frac{1}{2} (y - x)^T H_{\lambda^*(x)} (y - x) + O(\|y - x\|^3).$$

Multiplying both sides by $K(\|y - x\|)$ and integrating in y we have

$$\begin{aligned} \int_{B(x,r)} K(\|y - x\|) \lambda^*(y) dy &= \int_{B(x,r)} K(\|y - x\|) \lambda^*(x) dy + \int_{B(x,r)} K(\|y - x\|) \nabla \lambda^*(x)^T (y - x) dy \\ &\quad + \frac{1}{2} \int_{B(x,r)} K(\|y - x\|) (y - x)^T H_{\lambda^*(x)} (y - x) dy + O(r^5). \end{aligned}$$

We evaluate the three integrals from the Taylor expansion. Substituting $\tilde{y}_i = (y_i - x_i)/r$ in the first integral gives

$$\lambda^*(x) \int_{B(x,r)} K(\|y - x\|) dy = \lambda^*(x) r^2 \int_{B(0,1)} \bar{K}(\|\tilde{y}\|) d\tilde{y} = \lambda^*(x) r^2 q^{(1)}.$$

The linear term vanishes by symmetry of the ball. Indeed, substituting $\tilde{y}_i = (y_i - x_i)/r$:

$$\int_{B(x,r)} K(\|y - x\|) \nabla \lambda^*(x)^T (y - x) dy = r^3 \sum_{i=1}^2 \frac{\partial \lambda^*(x)}{\partial x_i} \int_{B(0,1)} \bar{K}(\|\tilde{y}\|) \tilde{y}_i d\tilde{y}.$$

Using polar coordinates in the first component gives

$$\int_0^{2\pi} \int_0^1 \bar{K}(\rho) \rho^2 \cos \phi d\rho d\phi = \int_0^1 \bar{K}(\rho) \rho^2 d\rho \int_0^{2\pi} \cos \phi d\phi = 0.$$

For the same reason, the second component also vanishes. For the third term, substituting $\tilde{y}_i = (y_i - x_i)/r$ gives

$$\frac{1}{2} \int_{B(x,r)} K(\|y - x\|) (y - x)^T H_{\lambda^*(x)} (y - x) dy = \frac{r^4}{2} \int_{B(0,1)} \bar{K}(\|\tilde{y}\|) \tilde{y}^T H_{\lambda^*(x)} \tilde{y} d\tilde{y}.$$

We can write $\tilde{y}^T H_{\lambda^*(x)} \tilde{y} = \sum_{i,j \in \{1,2\}} \frac{\partial^2 \lambda^*(x)}{\partial x_i \partial x_j} \tilde{y}_i \tilde{y}_j$ and using polar coordinates we observe that $\int_{B(0,1)} \bar{K}(\|\tilde{y}\|) \tilde{y}_i \tilde{y}_j d\tilde{y}$ vanishes when $i \neq j$ (since $\int_0^{2\pi} \cos \phi \sin \phi d\phi = 0$) and equals $q^{(3)}/2$ otherwise (since $\int_0^{2\pi} \cos^2 \phi d\phi = \int_0^{2\pi} \sin^2 \phi d\phi = \pi$). Therefore:

$$\frac{1}{2} \int_{B(x,r)} K(\|y - x\|) (y - x)^T H_{\lambda^*(x)} (y - x) dy = \frac{r^4 q^{(3)}}{4} \nabla^2 \lambda^*(x).$$

In addition, by the symmetry of the ball, the coefficient of r^5 also vanishes.

Recall that $z^*(x) = \int_{B(x,r)} K(\|y - x\|) \lambda^*(y) dy$, so

$$z^*(x) = \lambda^*(x) r^2 q^{(1)} + \frac{r^4 q^{(3)}}{4} \nabla^2 \lambda^*(x) + O(r^6).$$

In a canonical discrete model with equal police capacity and budget (i.e., $\theta = r^2 q^{(1)}$), we have $z^* = r^2 q^{(1)} \lambda_0^*$, and therefore

$$\lambda_0^*(x) = \lambda^*(x) + \frac{r^2 q^{(3)}}{4 q^{(1)}} \nabla^2 \lambda^*(x) + O(r^4).$$

Observe that $\lambda^*(x) = \lambda_0^*(x) + O(r^2)$ (which follows from the expansion above), so $\nabla^2 \lambda^*(x) = \nabla^2 \lambda_0^* + O(r^2)$, and substituting:

$$\lambda_0^*(x) = \lambda^*(x) + \frac{r^2 q^{(3)}}{4 q^{(1)}} \nabla^2 \lambda_0^*(x) + O(r^4),$$

and thus

$$\lambda^*(x) - \lambda_0^*(x) = -\frac{r^2 q^{(3)}}{4 q^{(1)}} \nabla^2 \lambda_0^*(x) + O(r^4).$$

Define $\eta := q^{(3)}/(4q^{(1)})$ and the proof is completed. $\square$

Proof of Corollary 3

Proof. Taking the gradient of both sides in Eq. (8) to obtain

$$\nabla \lambda^*(x) = \nabla \lambda_0^*(x) - \eta r^2 \nabla (\nabla^2 \lambda_0^*) (x) + O(r^4).$$

Evaluating at $x^c = x_0^c + x_\epsilon$ and expanding around x_0^c gives

$$0 = \nabla \lambda_0^*(x_0^c) + H_{\lambda_0^*}(x_0^c) x_\epsilon - \eta r^2 \nabla (\nabla^2 \lambda_0^*) (x_0^c) + O(r^4),$$

where we used that $\nabla (\nabla^2 \lambda_0^*)(x_0^c + x_\epsilon) = \nabla (\nabla^2 \lambda_0^*)(x_0^c) + O(r^2)$ since $x_\epsilon = O(r^2)$. Since $\nabla \lambda_0^*(x_0^c) = 0$, this simplifies to

$$H_{\lambda_0^*}(x_0^c) x_\epsilon = \eta r^2 \nabla (\nabla^2 \lambda_0^*) (x_0^c) + O(r^4).$$

Inverting $H_{\lambda_0^*}(x_0^c)$, which is non-singular by assumption, yields the result. $\square$

Proof of Proposition 4

Proof. We have that $f_{\lambda^*} = \lambda^*/\Lambda$ and $f_{\lambda_0^*} = \lambda_0^*/\Lambda$. From $\lambda^*(x) = \lambda_0^*(x) - \eta r^2 \nabla^2 \lambda_0^*(x) + O(r^4)$:

$$\begin{aligned} \ln f_{\lambda^*}(x) &= \ln \frac{\lambda_0^*(x) - \eta r^2 \nabla^2 \lambda_0^*(x)}{\Lambda} \\ &= \ln f_{\lambda_0^*}(x) + \ln \left(1 - \frac{\eta r^2 \nabla^2 \lambda_0^*(x)}{\lambda_0^*(x)} \right) \\ &= \ln f_{\lambda_0^*}(x) - \frac{\eta r^2 \nabla^2 \lambda_0^*(x)}{\lambda_0^*(x)} + O(r^4), \end{aligned}$$

where the last step uses $\ln(1 - \epsilon) = -\epsilon + O(\epsilon^2)$ with $\epsilon = \eta r^2 \nabla^2 \lambda_0^*(x)/(\lambda_0^*(x)) = O(r^2)$.

Therefore

$$\begin{aligned} f_{\lambda^*}(x) \ln f_{\lambda^*}(x) &= \left(f_{\lambda_0^*}(x) - \frac{\eta r^2 \nabla^2 \lambda_0^*(x)}{\Lambda} + O(r^4) \right) \left(\ln f_{\lambda_0^*}(x) - \frac{\eta r^2 \nabla^2 \lambda_0^*(x)}{\lambda_0^*(x)} + O(r^4) \right) \\ &= f_{\lambda_0^*}(x) \ln f_{\lambda_0^*}(x) - \frac{1}{\Lambda} \eta r^2 \nabla^2 \lambda_0^*(x) - \frac{1}{\Lambda} \eta r^2 \nabla^2 \lambda_0^*(x) \ln f_{\lambda_0^*}(x) + O(r^4). \end{aligned}$$

Multiply by minus one and integrate in x in both sides to obtain:

$$\mathcal{H}(f_{\lambda^*}) = \mathcal{H}(f_{\lambda_0^*}) + \frac{\eta r^2}{\Lambda} \int \nabla^2 \lambda_0^*(x) dx + \frac{\eta r^2}{\Lambda} \int_W \nabla^2 \lambda_0^*(x) \ln f_{\lambda_0^*}(x) dx + O(r^4).$$

Since $\lambda_0^*(x)$ is an increasing transformation of $a(x)$, $\nabla a(x) \cdot \mathbf{n}(x) = 0$ implies that $\nabla \lambda_0^*(x) \cdot \mathbf{n}(x) = 0$. Using, in addition, the divergence theorem, we obtain $\int_W \nabla^2 \lambda_0^*(x) dx = \int_{\text{Bd}(W)} \nabla \lambda_0^*(x) \cdot \mathbf{n}(x) d\sigma(x) = 0$. Thus,

$$\mathcal{H}(f_{\lambda^*}) = \mathcal{H}(f_{\lambda_0^*}) + \frac{\eta r^2}{\Lambda} \int_W \nabla^2 \lambda_0^*(x) \ln f_{\lambda_0^*}(x) dx + O(r^4).$$

Applying Green's first identity:

$$\int_W \nabla^2 \lambda_0^*(x) \ln f_{\lambda_0^*}(x) dx = \int_{\text{Bd}(W)} \ln f_{\lambda_0^*}(x) \nabla \lambda_0^* \cdot \mathbf{n}(x) d\sigma(x) - \int_W \nabla \lambda_0^*(x) \cdot \nabla \ln f_{\lambda_0^*}(x) dx,$$

and from the boundary condition the first term in the right-hand side vanishes. Using the fact that $\nabla \ln f_{\lambda_0^*} = \nabla \lambda_0^* / \lambda_0^*$, we conclude the proof. $\square$

Proof of Claim 1

Proof. For a 2×2 symmetric matrix A , let $\|A\| = \sup_{\|\mathbf{v}\|=1} |\mathbf{v}^T A \mathbf{v}|$ and $A > 0$ if A is positive definite. A peak reversal occurs when x_0^c is a local maximum of λ_0^* and a local minimum of λ^* . Taking the Hessian in both sides of Eq. (8) and evaluating at x_0^c yields:

$$H_{\lambda^*}(x_0^c) = H_{\lambda_0^*}(x_0^c) - \eta r^2 H_{\nabla^2 \lambda_0^*}(x_0^c) + O(r^4). \quad (11)$$

Since $H_{\lambda_0^*}(x_0^c) < 0$, peak reversal requires $H_{\lambda^*}(x_0^c) \geq 0$. Let $\mathbf{e}$ be the unit eigenvector of $H_{\lambda_0^*}(x_0^c)$ corresponding to its most negative eigenvalue. If there is peak reversal, we must have $H_{\lambda^*}(x_0^c) \geq 0$, or, equivalently, to second order

$$0 \leq \mathbf{e}^T H_{\lambda_0^*}(x_0^c) \mathbf{e} - \eta r^2 \mathbf{e}^T H_{\nabla^2 \lambda_0^*}(x_0^c) \mathbf{e}.$$

Now we consider two cases for the sign of $\mathbf{e}^T H_{\nabla^2 \lambda_0^*}(x_0^c) \mathbf{e}$. If it is non-negative, peak reversal cannot occur since the right-hand side above is negative. If, on the other hand, $\mathbf{e}^T H_{\nabla^2 \lambda_0^*}(x_0^c) \mathbf{e} < 0$, then we can write

$$0 \leq \mathbf{e}^T H_{\lambda_0^*}(x_0^c) \mathbf{e} - \eta r^2 \mathbf{e}^T H_{\nabla^2 \lambda_0^*}(x_0^c) \mathbf{e} = -\|H_{\lambda_0^*}(x_0^c)\| + \eta r^2 |\mathbf{e}^T H_{\nabla^2 \lambda_0^*}(x_0^c) \mathbf{e}|,$$

where $\mathbf{e}^T H_{\lambda_0^*}(x_0^c) \mathbf{e} = -\|H_{\lambda_0^*}(x_0^c)\|$ since $\mathbf{e}$ is the eigenvector of the most negative eigenvalue of a negative definite matrix. This implies that

$$r^2 \geq \frac{1}{\eta} \frac{\|H_{\lambda_0^*}(x_0^c)\|}{|\mathbf{e}^T H_{\nabla^2 \lambda_0^*}(x_0^c) \mathbf{e}|} \geq \frac{1}{\eta} \frac{\|H_{\lambda_0^*}(x_0^c)\|}{\|H_{\nabla^2 \lambda_0^*}(x_0^c)\|}.$$

However, the second-order approximation in Eq. (11) is guaranteed to determine the sign of $H_{\lambda^*}(x_0^c)$ only when the retained correction is subdominant, that is, when

$$\eta r^2 \|H_{\nabla^2 \lambda_0^*}(x_0^c)\| \ll \|H_{\lambda_0^*}(x_0^c)\| \Leftrightarrow r^2 \ll \frac{1}{\eta} \frac{\|H_{\lambda_0^*}(x_0^c)\|}{\|H_{\nabla^2 \lambda_0^*}(x_0^c)\|}.$$

Peak reversal requires the opposite ordering: at such radii the retained correction is no longer subdominant, and the neglected $O(r^4)$ remainder is of the same order as the terms determining the sign. Therefore, whenever peak reversal occurs within the second-order approximation, the approximation's error is no longer guaranteed to be negligible, so the approximation cannot certify the reversal. $\square$

Proof of Claim 2

Proof. Since λ_0^* is radially symmetric with a local maximum at the origin, we can write $\lambda_0^*(x) = a_4\|x\|^4 + a_2\|x\|^2 + a_0$. Since λ_0^* is a polynomial of degree four, the fourth-order approximation is exact. The radial symmetry of λ_0^* and the crime-detecting kernel imply that λ^* is also radially symmetric. The Hessian of λ^* at the origin is given by

$$H_{\lambda^*}(0) = H_{\lambda_0^*}(0) - \eta r^2 H_{\nabla^2 \lambda_0^*}(0) = 2a_2 I - 32\eta r^2 a_4 I = 2(a_2 - 16\eta r^2 a_4) I.$$

Therefore, the matrix $H_{\lambda^*}(0)$ is positive definite if and only if $r^2 > a_2/(16\eta a_4) = \bar{r}^2$, i.e., $r > \bar{r}$. In this case, the origin becomes a local minimum of λ^* , concluding the peak reversal part of the proof.

The local maxima of λ^* satisfy $\nabla \lambda^*(x^c) = 0$. Using the fourth-order expansion of λ^* , $\lambda^*(x) = a_4\|x\|^4 + (a_2 - 16\eta r^2 a_4)\|x\|^2 + \text{const}$, we have

$$\nabla \lambda^*(x) = 2[2a_4\|x\|^2 + (a_2 - 16\eta r^2 a_4)]x.$$

For $x^c \neq 0$, the condition $\nabla \lambda^*(x^c) = 0$ gives

$$2a_4\|x^c\|^2 + (a_2 - 16\eta r^2 a_4) = 0,$$

yielding $\|x^c\|^2 = -(a_2 - 16\eta r^2 a_4)/(2a_4) = 8\eta(r^2 - \bar{r}^2)$, hence $\|x^c\| = \sqrt{8\eta(r^2 - \bar{r}^2)}$. The Hessian of λ^* at x^c is

$$H_{\lambda^*}(x^c) = 2[2a_4\|x^c\|^2 + (a_2 - 16\eta r^2 a_4)]I + 8a_4 x^c (x^c)^T = 8a_4 x^c (x^c)^T.$$

The eigenvalue in the radial direction is $8a_4\|x^c\|^2 = -4(a_2 - 16\eta r^2 a_4) < 0$ for $r > \bar{r}$, so $\|x^c\| = \sqrt{8\eta(r^2 - \bar{r}^2)}$ defines the locus of the local maxima of λ^* . $\square$

C Properties of PPPs Used in the Paper

We describe in the following a few other properties of PPPs that we use in this paper. Let Y be a PPP in W with intensity χ .

Intensity Defines the Distribution Law. The probability law of Y is entirely characterized by its intensity measure χ .

Characterization of the Process in Subsets. The set of points of Y lying in a set $A \subset W$ (i.e., the point process $A \cap Y$) inherits the Poisson properties, being a PPP with intensity $\chi(\cdot)\mathbb{1}_{\{(\cdot) \in A\}}$ and mean $\int_A \chi(x)dx$.

The Thinning Property. We call *thinning property* the fact that if each point of Y is removed independently and with a probability p_{th} (that might depend on the location in W), the remaining set of points is a PPP with intensity $(1 - p_{\text{th}})\chi$.

The Generating Functional. For any measurable function f , $\mathbb{E} [\prod_{x \in Y} f(x)]$, called *generating functional* with respect to f , is equal to $\exp(-\int_W (1 - f(x))\chi(x)dx)$.

Notation Summary

Symbol	Description
<i>Sets and geometry</i>	
$W \subset \mathbb{R}^2$	Compact set representing the police jurisdiction (city), with C^1 boundary
$\text{Bd}(W)$	Boundary of W ; $\mathbf{n}(x)$ is the outward unit normal at $x \in \text{Bd}(W)$
W'	An arbitrary (two-dimensional) region, $W' \subset W$
$ W' $	Area (Lebesgue measure) of the region W'
$\{W_i\}_{i \in I}$	Collection of disjoint regions partitioning (a subset of) W , $I \subseteq \mathbb{N}$
$B(x, r)$	Ball of radius r centered at x
$\mathcal{C}_{++}$	Set of positive continuous functions on W
δ	Dirac delta function
f_v	Normalization of a non-negative integrable v : $f_v := v / \int_W v(x) \, dx$
<i>Primitives and fundamentals</i>	
Λ	Police budget; mean number of patrols (Poisson)
b	Penalty imposed on a criminal when a crime is cleared
N	Number of potential criminals
κ	Idiosyncratic opportunity cost of committing a crime
F_κ	CDF of the opportunity cost κ
$a(x)$	Crime attractiveness: expected gain from an uncleared crime at x
$a(W')$	Average attractiveness over W' : $\frac{1}{ W' } \int_{W'} a(x) \, dx$
$\tilde{a}$	Adjusted attractiveness, $\tilde{a} := a + b$
$\star$	Superscript denoting equilibrium values
<i>Clearance mechanism</i>	

Table A1 (continued)

Symbol	Description
$K(\cdot)$	Crime-detecting kernel; $K(\ x - y\)$ is the detection probability of a patrol at y for a crime at x
r	Coverage radius: $K(t) = 0$ for $t > r$
q	Effective coverage of a patrol, $q := \int_{\mathbb{R}^2} K(\ x\) dx$
θ	Discrete-model counterpart of q in the canonical discrete model
C	Police capacity, $C := \Lambda q/ W $ (or $\Lambda \theta/ W $ in a canonical discrete model)
$\pi_{W'}(\Lambda')$	Public safety production function (PSPF): clearance probability in W' given expenditure Λ'
<i>Point processes and intensities</i>	
Φ	Point process of crime locations
Ψ	Point process of patrol locations (a Poisson point process)
μ	Intensity of Φ : surface density of crimes (crime “heat map”)
λ	Intensity of Ψ : surface density of patrols (police allocation)
$z(x)$	Police activity at x : average number of patrols that would detect a crime at x
<i>Payoffs and equilibrium objects</i>	
$U^{\text{pol}}(\lambda, \mu)$	Expected utility of the police command (expected cleared crimes)
$U^{\text{crim}}(\lambda, s)$	Expected payoff of a criminal using location strategy s
$s, \mathcal{S}$	A criminal’s location strategy (distribution over W); $\mathcal{S}$ is the space of such distributions
$\underline{C}(f)$	Capacity threshold constant, $\frac{1}{ W } \int_W \ln f(x) dx - \ln \min\{f\}$
$M_{W'}^*$	Expected number of crimes in W' (discrete model)
$\Lambda_{W'}^*$	Equilibrium police expenditures in W' (discrete model)

Table A1 (continued)

Symbol	Description
$p_{W'}^*$	Expected proportion of cleared crimes in W' (aggregate clearance rate, ACR, discrete model)
V^*	Equilibrium value of crime
M^*	Equilibrium crime rate
<i>Information-theoretic quantities</i>	
u, U	Uniform distribution over W ; U a uniform random variable
$\mathcal{H}(f_1)$	Entropy of a density f_1 , $-\int_W f_1 \ln f_1 dx$
$\mathcal{H}(f_1, f_2)$	Cross-entropy, $-\int_W f_1 \ln f_2 dx$
$D(f_1 \ f_2)$	Relative entropy (Kullback–Leibler divergence), $\int_W f_1 \ln(f_1/f_2) dx$
$\mathcal{I}(f_1)$	Fisher information of f_1 , $\int_W \ \nabla f_1\ ^2 / f_1 dx$
<i>Spillover analysis</i>	
λ_0^*	Police intensity, no spillovers (canonical discrete model, $r \rightarrow 0$)
∇, ∇^2	Gradient and Laplacian operators
H_h	Hessian of a function $h : W \rightarrow \mathbb{R}$
$\eta, \tilde{\eta}$	Positive kernel-dependent constants (second- and fourth-order coefficients)
x_0^c, x^c	Critical point of λ_0^* and its counterpart for λ^*
x_ϵ	Displacement of a critical point induced by spillovers
$\bar{r}$	Peak-reversal threshold radius

Supplementary Online Material: Properties of the PSPF Induced by our Continuous Model

In this appendix, we show that adding two properties to consistent aggregation characterizes the PSPF induced by our model: adopting this PSPF in a discrete model is equivalent to imposing the three properties. Since these properties are intuitive, our continuous model provides a principled foundation for discrete-choice models of crime, singling out a well-behaved PSPF to work with. The first property is the consistent aggregation principle.

The second property we impose on models of crime is that if the surface density of enforcement around a point—defined as the ratio of police expenditures to area in a region—converges to zero, then the ACR around that point must also converge to zero. This captures the idea that, in the absence of police enforcement in the vicinity of a point, no crimes committed there can be cleared. We formalize this requirement as follows and we call it the NENC principle.

No-Enforcement, No-Clearance Principle. *For any sequence $\{W'_i\}_i$ of subsets of W such that $W'_i \rightarrow \{x\}$ (under the Hausdorff distance) and any sequence of expenditures $\{\Lambda_{W'_i}^*\}_i$ satisfying $\lim_{i \rightarrow \infty} \Lambda_{W'_i}^*/|W'_i| = 0$, it follows that $\lim_{i \rightarrow \infty} p_{W'_i}^* = 0$.*

The third property we impose on models of crime is that the cost of raising the clearance rate in a region depends on the current clearance rate and the size of the region, but not on how attractive the region is to criminals. This captures the idea that the PSPF is a description of police technology—how enforcement expenditures convert into cleared crimes—and should therefore not be calibrated to criminal preferences: two regions of the same size and the same clearance rate require the same additional expenditure to achieve the same improvement, regardless of the gains criminals derive from acting there. We formalize this requirement as follows and we call it the AI principle.

Attractiveness Independence Principle. *For any two regions W' and W'' with $|W'| = |W''|$, and any clearance levels u_1, u_2 , the additional expenditure required to raise the clearance rate from u_1 to u_2 is the same in W' and W'' . That is,*

$$\pi_{W'}^{-1}(u_2) - \pi_{W'}^{-1}(u_1) = \pi_{W''}^{-1}(u_2) - \pi_{W''}^{-1}(u_1).$$

Proposition 5. *The PSPF given by Eq. (7) is the unique PSPF, up to a scale parameter,*

allowing a discrete-choice model to respect simultaneously the consistent aggregation, NENC, and AI principles.

Proof. We prove it in three steps, using each property at each step.

First step: consistent aggregation. Fix a partition and let V^* denote the value of committing a crime, so that criminal indifference across cells gives $\tilde{a}(W')(1 - u_{W'}) - b = V^*$, where $u_{W'} = \pi_{W'}(\Lambda_{W'})$. The equalization conditions across any two cells imply that, at each equilibrium,

$$\frac{\pi_{W'}^{-1'}(1 - (V^* + b)/\tilde{a}(W'))}{|W'| \tilde{a}(W')} = c$$

for a constant c common across cells. Crucially, c may depend on the equilibrium: $c = c(V^*)$. Substituting $u = 1 - (V^* + b)/\tilde{a}(W')$ gives the following functional equation

$$\pi_{W'}^{-1'}(u) = |W'| \tilde{a}(W') c(\tilde{a}(W')(1 - u) - b).$$

Let Γ be an antiderivative of c (strictly increasing, since $c > 0$). Integrating in u ,

$$\pi_{W'}^{-1}(u) = H(W') - |W'| \Gamma(\tilde{a}(W')(1 - u) - b),$$

where $H(W')$ is a region-specific integration constant. Consistent aggregation pins down its structure: in equilibrium, $\Lambda_{W'} = H(W') - |W'| \Gamma(V^*)$, so for any cell W' partitioned into W_1 and W_2 , invariance of the equilibrium outcomes (V^* and the total budget) requires

$$H(W_1) + H(W_2) - (|W_1| + |W_2|) \Gamma(V^*) = H(W') - |W'| \Gamma(V^*),$$

hence H is additive: $H(W_1 \cup W_2) = H(W_1) + H(W_2)$. By additivity and absolute continuity, $H(W') = \int_{W'} h(x) dx$ for some density h . Therefore, consistent aggregation restricts the PSPF to the family

$$\pi_{W'}^{-1}(u) = \int_{W'} h(x) dx - |W'| \Gamma(\tilde{a}(W')(1 - u) - b), \quad (12)$$

indexed by the pair (Γ, h) .

Second step: the NENC principle. Take a sequence of cells $\{W_n\}_n$ with $\max_n \text{diam}(W_n) \rightarrow 0$ shrinking to an arbitrary point $x \in W$, and budgets with $\Lambda_n/|W_n| \rightarrow 0$. Evaluating Eq. (12) at the equilibrium clearance $u_n = \pi_{W_n}(\Lambda_n)$ and dividing by $|W_n|$,

$$\Gamma(\tilde{a}(W_n)(1 - u_n) - b) = \frac{1}{|W_n|} \int_{W_n} h(x') dx' - \frac{\Lambda_n}{|W_n|}.$$

As $n \rightarrow \infty$: the last term vanishes; by the Lebesgue differentiation theorem (using $\text{diam}(W_n) \rightarrow 0$), the first term on the right converges to $h(x)$ for almost every x ; by continuity of a , $\tilde{a}(W_n) \rightarrow \tilde{a}(x)$; and the NENC principle requires $u_n \rightarrow 0$. By continuity of Γ ,

$$\Gamma(\tilde{a}(x) - b) = h(x) \quad \text{for a.e. } x \in W.$$

Substituting into Eq. (12), the family reduces to a single functional parameter:

$$\pi_{W'}^{-1}(u) = \int_{W'} \Gamma(\tilde{a}(x) - b) dx - |W'| \Gamma(\tilde{a}(W')(1 - u) - b). \quad (13)$$

Third step: the AI principle. By the attractiveness independence principle, for any two regions W' and W'' with $|W'| = |W''|$ and any clearance levels u_1, u_2 , the additional expenditure required to raise the clearance rate from u_1 to u_2 coincides:

$$\pi_{W'}^{-1}(u_2) - \pi_{W'}^{-1}(u_1) = \pi_{W''}^{-1}(u_2) - \pi_{W''}^{-1}(u_1).$$

Using Eq. (13), and noting that the additive term $\int_{W'} \Gamma(\tilde{a}(x) - b) dx$ cancels in the difference $\pi_{W'}^{-1}(u_2) - \pi_{W'}^{-1}(u_1)$, this expenditure is

$$\pi_{W'}^{-1}(u_2) - \pi_{W'}^{-1}(u_1) = |W'| \left[\Gamma(\tilde{a}(W')(1 - u_1) - b) - \Gamma(\tilde{a}(W')(1 - u_2) - b) \right].$$

Since $|W'| = |W''|$, the AI principle requires

$$\Gamma(\tilde{a}(W')(1 - u_1) - b) - \Gamma(\tilde{a}(W')(1 - u_2) - b) = \Gamma(\tilde{a}(W'')(1 - u_1) - b) - \Gamma(\tilde{a}(W'')(1 - u_2) - b)$$

for all u_1, u_2 and all admissible values $\tilde{a}(W'), \tilde{a}(W'')$. Hence the map $t \mapsto \Gamma(t(1 - u_1) - b) - \Gamma(t(1 - u_2) - b)$ is constant in t ; differentiating in t gives

$$(1 - u_1) \Gamma'(t(1 - u_1) - b) = (1 - u_2) \Gamma'(t(1 - u_2) - b),$$

so the map $u \mapsto (1-u)\Gamma'(t(1-u)-b)$ is constant. Writing $V+b=t(1-u)$, this reads $(V+b)\Gamma'(V) = \text{const}$, whose solutions are

$$\Gamma(V) = \frac{1}{\theta} \ln(V+b) + \text{const}$$

for any $\theta > 0$. Substituting $\Gamma = \frac{1}{\theta} \ln(\cdot + b)$ into Eq. (13) and inverting,

$$\pi_{W'}(\Lambda') = 1 - \frac{e^{\frac{1}{|W'|} \int_{W'} \ln \tilde{a}(x) dx}}{\tilde{a}(W')} e^{-\theta \Lambda' / |W'|} = 1 - e^{-D(u \| f_{\tilde{a}, W'})} e^{-\theta \Lambda' / |W'|},$$

which is the PSPF generated by the topological model. Conversely, this PSPF belongs to the family (12), satisfies the NENC principle (as $\text{diam}(W') \rightarrow 0$, the divergence vanishes), and satisfies the AI principle by the computation above. Hence it is the unique PSPF, up to the scale parameter θ , respecting consistent aggregation, the NENC principle, and the AI principle. $\square$